

**AN
INTERNSHIP REPORT
On**

Gym Finder & Recommender

Submitted

by

**Aashish Haripara
(Enrollment Number: 211430116040)**

Guided By:

Prof. Divyang Makwana

An

Internship Report

Submitted to



Gujarat Technological University

In fulfillment for the award of degree of Bachelor of Engineering
in

INFORMATION TECHNOLOGY

ACADEMIC YEAR – 2025



NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

Pakwan, Behind Rajpath Club Gate to Sindhu Bhavan Road,
Sarkhej - Gandhinagar Hwy, AEC Char Rasta, Ahmedabad, Gujarat 380054



NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

CERTIFICATE

This is to certify that the Internship report submitted along with the project entitled Internship in **Impero IT services PVT LTD** has been Completed by **Aashish Haripara** under my guidance in complete fulfilment for the Bachelor of Engineering in **Information Technology** Branch, 8th Semester of **Gujarat Technological University**, Ahmedabad during the academic year 2025.

Date:

Place: NEW LJET, Ahmedabad.

Signature and Name of Guide

Prof. Divyang Makwana
Assistant Professor (IT),
IT Department,
NEW LJET (143), Ahmedabad.

Signature and Name of H.O.D.

Dr. Gayatri S. Pandi.
IT Department,
NEW LJET (143), Ahmedabad.

Signature and Name of Principal

Dr. Gayatri S. Pandi.
NEW LJET (143), Ahmedabad

Seal of Institute



GUJARAT TECHNOLOGICAL UNIVERSITY

CERTIFICATE FOR COMPLETION OF ALL ACTIVITIES AT ONLINE PROJECT PORTAL

B.E. SEMESTER VIII, ACADEMIC YEAR 2024-2025

Date of certificate generation : 12 April 2025 (10:11:05)

This is to certify that, **Haripara Aashishkumar Amrutbhai** (Enrolment Number - 211430116040) working on project entitled with **GYM Finder & Recommender** from **INFORMATION TECHNOLOGY** department of **New L.J Institute of Engineering and Technology** had submitted following details at online project portal.

Internship Project Report	Completed
---------------------------	-----------

Name of Student : Haripara Aashishkumar
Amrutbhai

Name of Guide : M r . B H A U M I K
HASMUKHBHAI GELANI
M r . D I V Y A N G
PRAHLADBHAI MAKWANA

Signature of Student : _____

*Signature of Guide : _____

Disclaimer :

This is a computer generated copy and does not indicate that your data has been evaluated. This is the receipt that GTU has received a copy of the data that you have uploaded and submitted as your project work.

*Guide has to sign the certificate, Only if all above activities has been Completed.

Internship Completion Certificate



Impero IT Services Pvt. Ltd.

You, We & IT[®]

Internship Completion Certificate

"Life is a journey with problems to solve, lessons to learn, but most of all, experiences to enjoy."

18 Apr, 2025

This is to certify that **Mr. Aashish Amrutbhai Haripara** has successfully completed his Internship with **Impero IT Services Pvt Ltd** as a **Software Engineer - Trainee** from **20 Jan, 2025** to **19 Apr, 2025**

During the period of the Internship program, he had been exposed to different processes and was found to be diligent, hardworking, and inquisitive.

He was cooperative and sincere during the complete working tenure.

We wish him All good luck and growth!

For, Impero IT Services Pvt Ltd

Julie Braganza
HR Manager



NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

DECLARATION

I hereby declare that the Internship report submitted along with the Internship entitled Project **GYM Finder & Recommender** submitted in Complete for Bachelor of Engineering in **IT** Branch to **Gujarat Technological University**, Ahmedabad, is a bonafide record of original Internship work Completed by me at **Impero IT services PVT LTD** under the supervision of **External Guide Julie Braganza** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of Student

Aashish Haripara

Signature of Student

ACKNOWLEDGEMENT

I wish to express my sincere gratitude to my **External guide Julie Braganza** for continuously guiding me at the company and answering all my doubts with patience. I would also like to thank **Dr. Gayatri S. Pandi (H.O.D. of IT Department)** for motivating me every time whenever I get confused, I would also like to thank my **Internal Guide Prof. Divyang Makwana** for helping me through my internship by giving me the necessary suggestions and advices along with their valuable co-ordination in completing this Internship.

I also thank my parents, friends and all the members of the family for their precious support and encouragement which they had provided in completion of my work. In addition to that, I would also like to mention the company personals who gave me the permission to use and experience the valuable resources required for the Internship.

Thus, in conclusion to the above said, I once again thank the staff members of **Impero IT Services PVT LTD** for their valuable support in completion of the Internship.

Thank You.

Name of The Student

Aashish Haripara

Enrollment Number

211430116040

Date:

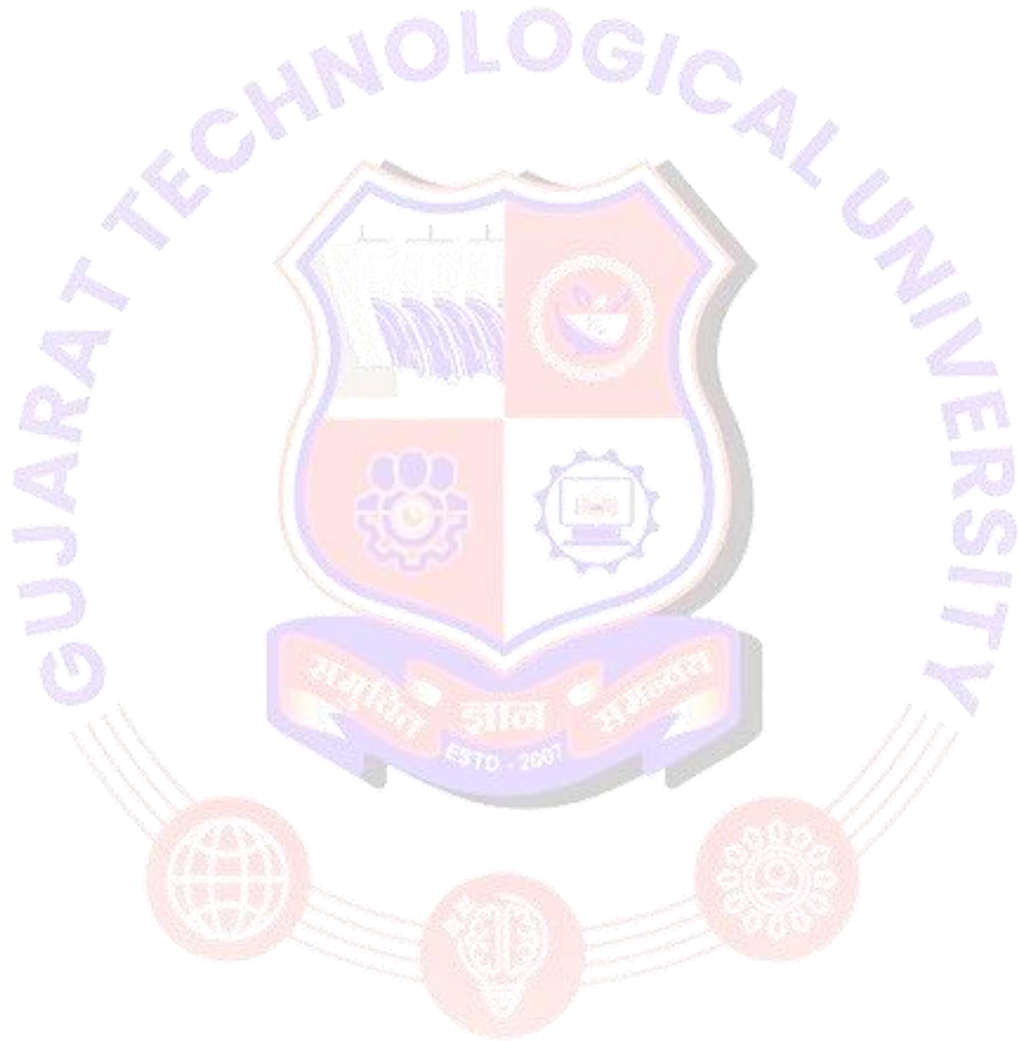
Signature of Student

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List of Abbreviations

HTML	HyperText Markup Language
CSS	Cascading Style Sheet
JS	Javascript
EJS	Embedded JavaScript
JWT	JsonWebToken

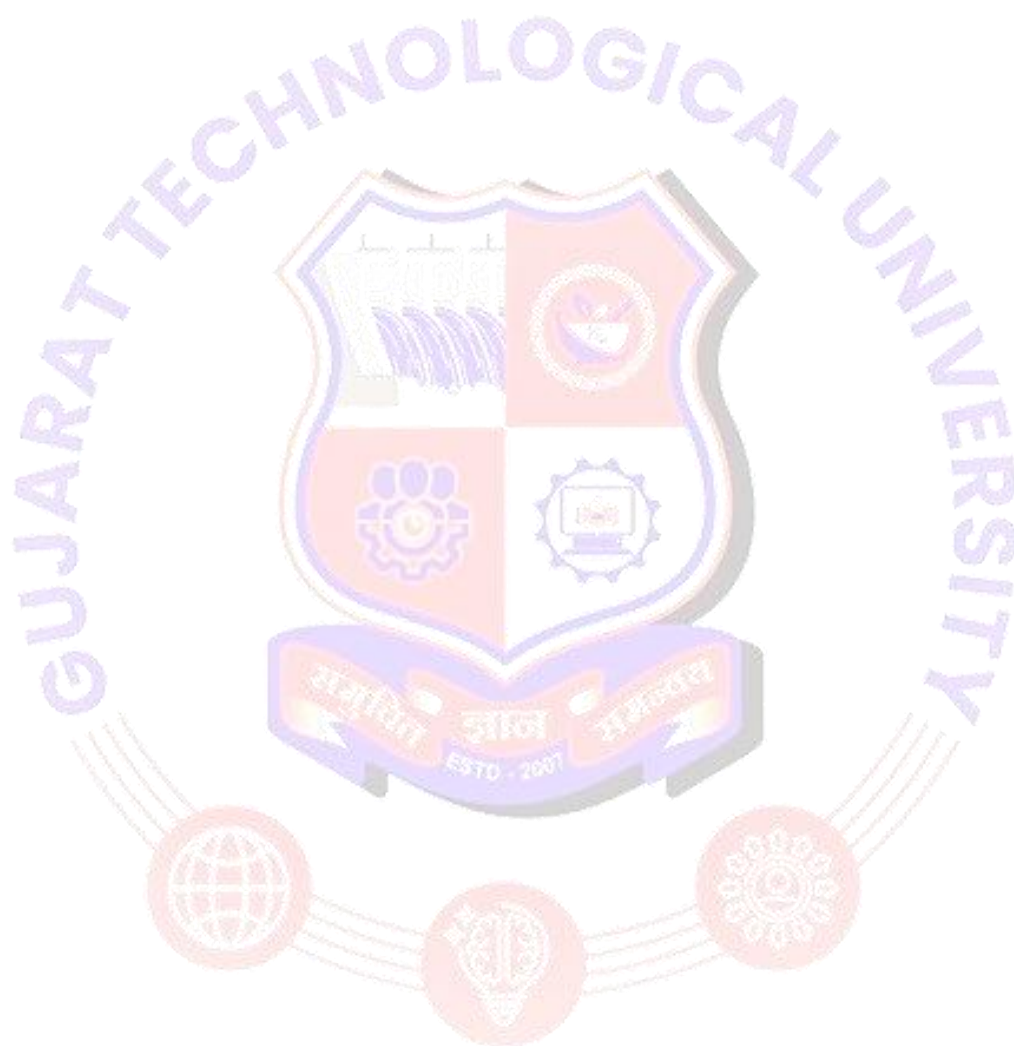


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GYM Finder & Recommender

Enrollment No.: 211430116040

Student Name: Aashish Haripara

**NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY
(College Code:143)**

Semester: VIII, IT Department

ABSTRACT

I am developing a GYM Finder & Recommender for the Company. Gym search and recommendations make choosing a gym easy by providing personalized recommendations based on user preferences, location, budget, and amenities. It collects and analyzes data from gyms within a certain radius to make accurate recommendations. Built on the MERN team and EJS standards, it integrates OAuth and JWT authentication for security. The system helps gym owners reach the right members while improving users' decision-making capabilities through location-based search, real-time data updates, and insights.

Keywords: *Gym, Gym-finder, Gym-recommender, Company, Working Strategy, Trainer, Priority, Status, Type, Group, History, Module*

CHAPTER: 1

INTRODUCTION OF PROJECT & COMPANY PROFILE

1.1. Introduction

1.1.1: Company Profile

1.1.2: Company Products

1.1.3: Company Mission and Vision

1.2. Introduction of the Project

1.2.1: Purpose of the Project

1.2.2: Function Requirements

1.2.3: Problems in Existing System

1.2.4: Main Modules

1.1. Introduction

1.1.1. Company Profile

We offer a full range of IT services encompassing web hosting, website development, mobile app development, MLM software consulting and development, e-commerce, as well logo and brochure designing and development.

Impero IT Services PVT LTD is a brain child of Mr. Raza Noorani who partnered to initiate the ambitious venture. Started as a small partnership firm way back in 2013, it grew exponentially within a decade.

The glorious journey that started with web development, MLM, and digital marketing services, moved on with new additions every year. The firm got its new structure as a company named Impero IT Services and now has grown into a full-fledged IT team of more than 110.

With its expansion, the company is all set to provide a comprehensive range of services to the global market. Having already worked on more than 30 projects and served top-notch clients from countries like Australia, Singapore, and Canada, we are expanding our horizons to the other untraded markets as well.

1.1.2. Company Products

- Impero IT Services PVT LTD is an Information Technology Training Institute, offering a wide area of solutions customized for a range of key verticals and horizontals in the information technologies segment. People with the knowledge of JavaScript and Node.js are highly sought after in the world today. If you want to excel at something that very few people are adept at, the Node.js course Ahmedabad is the right solution for you. This course offers in-depth information about JavaScript and Node which can help you establish a strong career in web development.
- ExpressJs is very well-known as it is powerful, straightforward, scalable, protected and highly customizable. Impero IT Services PVT LTD provides Node js and ExpressJS training to for enhancing website, adding new modules and

Plugins into existing website. Impero IT Services PVT LTD follows step-by-step approach when providing Nodejs Services to get affirmative results which will add unique needs.

- Impero IT Services PVT LTD is also providing Project Training for the implementation of the right tools for enterprise-wide planning and software development.
- Impero IT Services PVT LTD service portfolio encompasses the entire range of solutions required by the IT industry. We also offer placement services to fresher and experienced professionals in Ahmedabad.
- Impero IT Services PVT LTD is imparting high-quality education in the realm of major programming and non-programming skills including Node with ExpressJS development, Java, Python, ASP.Net, mobile app development, digital marketing, graphic designing, web designing, SEO, and many others.
- The learners get intensive coaching from the highly experienced trainers with best classroom environment and live project training.

1.1.3. Company Mission and Vision

- **Company Mission:**
 - To provide the most effective and friendly computer training which will result in productive and satisfied users of all levels.
 - Impero IT Services PVT LTD aims at delivering cutting-edge web solutions to the clients and helping them capitalize the optimum power of the IT technologies.
 - We ensure all our designing and development assignments are enhancing performance and revenues of your business.
 - Our training programs are specifically designed to train candidates to work in the industry as soon as they complete training.
- **Company Vision:**
 - As an established company now, we visualize our success in becoming a top-notch IT company and training Centre for a plethora of skills and technologies.
 - Keeping our customer's satisfaction at the Centre, we also provide the best platform for the fresh talents to learn, nurture, and grow.

1.2. Introduction of the Project

1.2.1. Purpose of the Project

- The aim of **gym search and recommendations** is to provide users with effective and personalized solutions to find gyms based on their **specific needs, location and budget**.
- The system aims to simplify the gym selection process by providing recommendations and detailed information about gyms in a specific area.
- The program aims to improve **user experience** and promote **healthy** living through better access to exercise by integrating features such as **adaptive changes, real-time data updates** and personalized recommendations.
- It also aims to empower gym owners by providing them with **a digital platform** to reach a wider audience and expand their membership base.

1.2.2. Function Requirements

- User Authentication & Authorization: Role based access for gym admin trainer or users.
- Location Based Gym finder functionality: You can easily get the perfect gym that fulfills your location requirements.
- Gym recommendation based on interest: You can find gym based on time, budget, gym-instruments, trainer facilities etc.
- Easy n-days booking features: Unlike traditional gyms with rigid monthly memberships, our system offers a flexible 10-day booking option, allowing users to try out different gyms without committing to long-term contracts or monthly charges.
- Secure payment integration: You can easily pay the membership fees with card, Online or net banking etc.

1.2.3. Problems in Existing System

- Limited Personalization: Most current systems cannot provide personalized recommendations based on a person's interests,
- Outdated or Incomplete information: Gym information like prices, equipment, trainers.

- Users are often forced to pay a monthly fees for no option of a good fit for their needs. Additionally, a lack of trial classes or safety measures limits user ability to try out gym services before making a long-term commitment.

1.2.4. Main Modules

- **Gym-owner Side**

- Manage The Gym.
- Edit the functionality
 - Add Gym
 - Delete Gym.
 - Update Gym
 - Changing the information like priority, Duration, Start date, End date
- Assign the Gym to Resources.
- Advance Search
 - By Gym Name
 - By Gym Location
- Show the Recent Booking.
- Show the List of Gyms and members.

- **Member Side**

- Member can Login Properly with Member Id & Password.
- Update the Status(active/inactive)
- Show the Recent Activity.
- Show the List of Tasks.
- Attach the Files of Diet Routine.

CHAPTER: 2

SYSTEM REQUIREMENTS

2.1 Hardware & Software Requirements

2.1.1: Server-Side Requirements

2.1.2: Developer Side Requirements

2.1.3: User Side Requirements

2.1 Hardware & Software Requirements

2.1.1 Server-Side Requirements

- **Hardware Requirement**
 - I3 Processor or above.
 - 500 GB HDD or above
 - 8GB of RAM.
 - Keyboard (Normal or Multimedia) and Mouse (Compatible Mouse)
- **Software Requirement**
 - **Front End:** HTML, CSS, EJS (Embedded JavaScript), and JavaScript
 - **Back End:** Node v18 or above and mongoDB
 - **Server:** MongoDB atlas
 - **Design Tool:** VS Code
 - **OS:** Window 7 or Any Compatible OS

2.1.2 Developer Side Requirements

- **Hardware Requirement**
 - I3 Processor or above.
 - 500 GB HDD or above
 - 8GB of RAM.
 - Keyboard (Normal or Multimedia) and Mouse (Compatible Mouse)
- **Software Requirement**
 - **Front End:** HTML, CSS, EJS (Embedded JavaScript), and JavaScript
 - **Back End:** Node v18 or above and mongoDB
 - **Server:** MongoDB atlas
 - **Design Tool:** VS Code
 - **OS:** Window 7 or Any Compatible OS
 -

2.1.3 User Side Requirements

- **Hardware Requirement**
 - I3 Processor or above.
 - GB HDD or above
 - 8GB of RAM.
 - Keyboard (Normal or Multimedia) and Mouse (Compatible Mouse)
- **Software Requirement**
 - OS: Window 7 or Any Compatible OS
 - **Browser:** Chrome or any Compatible browser
- **Basic Requirement**
 - Internet Connection.
 - **Front End:** EJS, JavaScript
 - **Back End:** Node v18 or above and mongoDB

CHAPTER: 3

WORK SHEET REPORT

3.1 Work Sheet Report (15 DAYS)

3.1 Work Sheet Report (15 DAYS):

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Aashish Haripara		
Enrollment No:		211430116040		
Internship/Project Title		GYM Finder & Recommender		
Tools and Technologies		• Front End: HTML, CSS, EJS and JS • Back End: NodeJS v18 and MongoDB • Server: MongoDB atlas • Design Tool: VS Code		
Company/ Organization Name		Impero IT services		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
1	20/01/2025 To 24/01/2025	- JavaScript Introduction - Basic to Advance JavaScript	Completed	
2	27/01/2025 To 31/01/2025	- JavaScript DOM Introduction - Project Tasks Based on DOM	Completed	

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Aashish Haripara		
Enrollment No:		211430116040		
Internship/Project Title		GYM Finder & Recommender		
Tools and Technologies		• Front End: HTML, CSS, EJS and JS • Back End: NodeJS v18 and MongoDB • Server: MongoDB atlas • Design Tool: VS Code		
Company/ Organization Name		Impero IT services		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
3	03/02/2025 To 07/02/2025	- Node.js Introduction - JS refresher - Understanding Basics of Node	Completed	
4	10/02/2025 To 14/02/2025	- Working With ExpressJs - Adding Templating Engine (EJS) - MVC - Routes and Models - SQL introduction	Completed	

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Aashish Haripara		
Enrollment No:		211430116040		
Internship/Project Title		GYM Finder & Recommender		
Tools and Technologies		• Front End: HTML, CSS, EJS and JS • Back End: NodeJS v18 and MongoDB • Server: MongoDB atlas • Design Tool: VS Code		
Company/ Organization Name		Impero IT services		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
5	17/02/2025 To 21/02/2025	- Understanding sequelize - Understanding MongoDB, Mongoose	Completed	
6	24/02/2025 To 28/02/2025	- Session and Cookies - Adding Authentication - Sending Email - Task using OAuth	Completed	

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Aashish Haripara		
Enrollment No:		211430116040		
Internship/Project Title		GYM Finder & Recommender		
Tools and Technologies		• Front End: HTML, CSS, EJS and JS • Back End: NodeJS v18 and MongoDB • Server: MongoDB atlas • Design Tool: VS Code		
Company/ Organization Name		Impero IT services		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
7	03/03/2025 To 07/03/2025	- Payment Integration Using Stripe - NodeJS Practice Task (LMS)	Completed	
8	10/03/2025 To 14/03/2025	- NodeJS Practice Task (Library Management System)	Completed	

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Aashish Haripara		
Enrollment No:		211430116040		
Internship/Project Title		GYM Finder & Recommender		
Tools and Technologies		• Front End: HTML, CSS, EJS and JS • Back End: NodeJS v18 and MongoDB • Server: MongoDB atlas • Design Tool: VS Code		
Company/ Organization Name		Impero IT services		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
9	17/03/2025 To 21/03/2025	- LMS - Model creation - View creation - Controller - Pagination - Authentication	Completed	
10	24/03/2025 To 28/03/2025	- Working on final project (Gym recommendation and finder)	Completed	

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Aashish Haripara		
Enrollment No:		211430116040		
Internship/Project Title		GYM Finder & Recommender		
Tools and Technologies		• Front End: HTML, CSS, EJS and JS • Back End: NodeJS v18 and MongoDB • Server: MongoDB atlas • Design Tool: VS Code		
Company/ Organization Name		Impero IT services		
Student’s Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
11	31/03/2025 To 04/04/2025	- Make Frontend Part of Gym recommendation - Make ejs templates for html views	Completed	
12	07/04/2025 To 11/04/2025	- Working on creating models and controllers add authentication and test whole	Completed	

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Aashish Haripara		
Enrollment No:		211430116040		
Internship/Project Title		GYM Finder & Recommender		
Tools and Technologies		• Front End: HTML, CSS, EJS and JS • Back End: NodeJS v18 and MongoDB • Server: MongoDB atlas • Design Tool: VS Code		
Company/ Organization Name		Impero IT services		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
13	14/04/2025 To 18/04/2025	- Working on project	Completed	

CHAPTER: 4

FRONT END OF SYSTEM

4.1 About Front End

4.1.1: about HTML and CSS

4.1.2: about JavaScript

4.1.3: about EJS (Embedded JavaScript)

4.1 About Front End

- Front end development refers to the part of web development that focuses on creating the user interface and experience of a website or web application.
- It involves implementing designs provided by UI/UX designers using languages such as HTML, CSS, and JavaScript.

4.1.1 about HTML & CSS

- HTML stands for Hyper Text Markup Language.
- HTML describes the structure of a Web page.
- HTML consists of a series of elements.
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.
- **HTML Page Structure**

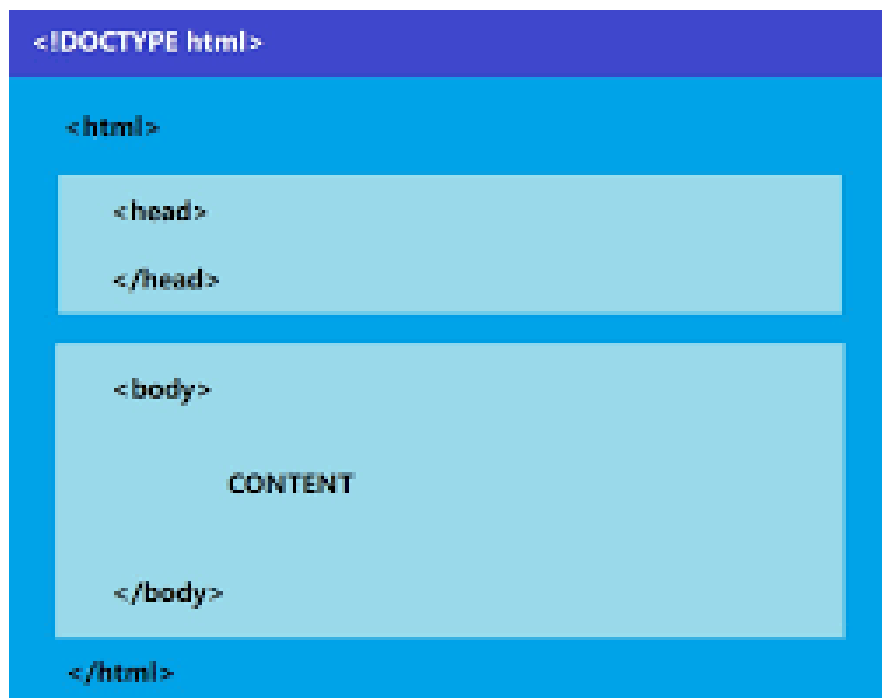


Fig. 4.1.1 HTML Page Structure

- Any HTML document mainly comprises two parts, the head and the body.

- The Head includes the additional information of the document, apart from the one displayed on the web page. It includes the HTML version, the title, and also the metadata.
- And the information that will be displayed on the web page is put inside the Body tag.
- HTML, or Hypertext Markup Language, is the standard language used to create and design web pages.
- It provides the structure and content of a webpage by using a system of markup tags and attributes.
- **Here's a breakdown of HTML's key components and concepts:**
 - **Elements:**
 - HTML documents are composed of elements, which are represented by tags enclosed in angle brackets (< and >).
 - Elements can have attributes that provide additional information about them.
 - **Tags:**
 - Tags are used to define the beginning and end of an element.
 - Some elements are self-closing, meaning they don't have content and close themselves within the opening tag.
 - **Attributes:**
 - Attributes provide additional information about an element and are specified within the opening tag.
 - They are comprised of a name and a value, separated by an equals sign (=).
 - **Document Structure:**
 - HTML documents have a specific structure defined by a hierarchy of elements.
 - The basic structure typically includes a <!DOCTYPE> declaration, <html>, <head>, and <body> elements.

- These elements include headings (<h1> to <h6>), paragraphs (<p>), lists (, ,), and more.
- **Links and Images:**
 - HTML allows you to include hyperlinks (<a>) to navigate between pages or sections within a page, and images () to display graphics on a webpage.
- HTML forms the backbone of web development, providing the structure and content for websites and web applications.
- It is often combined with CSS for styling and JavaScript for interactivity to create rich, dynamic web experiences.

about CSS

- CSS stands for Cascading Style Sheets.
- CSS describes how HTML elements are to be displayed on screen, paper, or in other media
- CSS saves a lot of work. It can control the layout of multiple web pages all at once
 - External stylesheets are stored in CSS files
- CSS, or Cascading Style Sheets, is a style sheet language used to describe the presentation of a document written in HTML or XML (including XML dialects like SVG or XHTML). CSS describes how elements should be rendered on screen, in print, or even spoken aloud.
 - **Here's a deeper look at CSS:**
 - **Selectors:**
 - CSS selectors are patterns used to select the elements you want to style.
 - They can target HTML elements, classes, IDs, attributes, and more.
 - **Properties and Values:**
 - CSS properties are the stylistic attributes you can apply to selected elements, and values determine how those properties are applied

- Units and Values:
 - CSS supports various units for specifying sizes, including pixels (px), percentages (%), ems (em), and more.
 - It also supports different value types such as colors, lengths, and keywords.
- **Box Model:**
 - CSS treats each HTML element as a box, comprising content, padding, border, and margin.
 - Understanding the box model is crucial for layout and positioning of elements on a webpage.
- **Responsive Design:**
 - CSS allows developers to create responsive designs that adapt to different screen sizes and devices using techniques like media queries, flexbox, and CSS grid.
- **Preprocessors and Frameworks:**
 - CSS preprocessors like Sass and Less extend the capabilities of CSS by adding features like variables, mixins, and nested rules.
 - CSS frameworks like Bootstrap and Foundation provide pre-written CSS and HTML templates to streamline development.
- CSS plays a fundamental role in web development by controlling the look and feel of web pages, making them visually appealing and user-friendly. It works hand-in-hand with HTML and often JavaScript to create dynamic and interactive web experiences.

4.1.2 about JavaScript

- JavaScript is a high-level, versatile programming language primarily known for its use in web development.
- It enables developers to add interactivity, behavior, and dynamic features to websites and web applications. Here's a comprehensive overview of JavaScript:

- **Client-Side Scripting:**

- JavaScript is primarily used as a client-side scripting language, meaning it runs on the user's web browser rather than a remote server.
- This allows for dynamic updates and interaction without needing to reload the entire webpage.

- **Core Features:**

- **Variables and Data Types:**

- JavaScript supports various data types such as numbers, strings, Booleans, arrays, objects, functions, and more.
- Variables are used to store data, and they can be dynamically typed.

- **Control Structures:**

- JavaScript provides control structures like conditionals (if, else if, else), loops (for, while, do-while), and switch statements.

- **Functions:**

- Functions are reusable blocks of code that perform a specific task.
- JavaScript supports both function declarations and function expressions, as well as arrow functions introduced in ES6.

- **Objects and Prototypes:**

- JavaScript is an object-oriented language, where objects are collections of key-value pairs.
- Objects can also inherit properties and methods from other objects through prototypes.

- **DOM Manipulation:**

- The Document Object Model (DOM) represents the structure of HTML documents as a tree of objects.
- JavaScript can manipulate the DOM to dynamically change the content, structure, and style of web pages.

- **Event-Driven Programming:**

- JavaScript follows an event-driven programming paradigm, where actions or events (like mouse clicks, key presses, or page loads) trigger the execution of specific code.

- Event handlers are used to respond to these events and perform actions accordingly.
- **Asynchronous Programming:**
 - JavaScript supports asynchronous programming through mechanisms like callbacks, promises, and async/await.
 - This allows non-blocking execution of code, enabling tasks like fetching data from servers, handling user input, and performing animations without freezing the UI.
- **Libraries and Frameworks:**
 - JavaScript has a vast ecosystem of libraries and frameworks that extend its capabilities and streamline development. Popular libraries include jQuery for DOM manipulation and AJAX requests, while frameworks like React.js, Angular, and Vue.js provide structured approaches to building complex web applications.
- **Server-Side Development:**
 - While JavaScript is primarily associated with client-side scripting, it is also used for server-side development.
 - Node.js is a runtime environment that allows JavaScript to be executed on the server, enabling full-stack JavaScript development.
- **Security Considerations:**
 - JavaScript runs in a sandboxed environment within the browser, which limits its access to the user's system.
 - However, developers must still be cautious of security vulnerabilities like cross-site scripting (XSS) and ensure proper data validation and sanitization.
- JavaScript is a fundamental technology for web development, empowering developers to create interactive and engaging web experiences across various devices and platforms.
- Its versatility and extensive ecosystem make it a cornerstone of modern web development.

4.1.3 about EJS (Embedded JavaScript)

- EJS, or Embedded JavaScript, is a lightweight templating engine used with Node.js to generate HTML markup with plain JavaScript.
- It enables developers to create dynamic web pages by embedding JavaScript code directly within HTML templates. EJS is known for its simplicity, flexibility, and ease of integration with Express.js, making it a popular choice for server-side rendering in Node.js-based web applications.
- With EJS, developers can pass data from the server to the front-end and display it dynamically, allowing for highly responsive and data-driven UI rendering.
- The syntax is clean and intuitive, using `<%= %>` for outputting variables and `<% %>` for control flow like loops and conditionals. This approach keeps HTML readable while still allowing powerful scripting capabilities.
- EJS also supports partials and layouts, allowing code reuse and modular design. This helps in maintaining a cleaner structure for larger applications. It does not require additional compilation steps and renders pages quickly on the server.
- **Key Features of EJS:**
 - Embeds JavaScript logic directly within HTML templates
 - Simple syntax for outputting data and handling control structures
 - Supports partials for reusable components (like headers/footers)
 - Easily integrates with Express.js and other Node.js frameworks
 - Useful for generating dynamic HTML content on the server
 - Lightweight with no need for a front-end framework
 - Enables server-side rendering with minimal overhead
- In this project, EJS is used to render views such as login pages, profile pages, and dashboards, allowing the application to dynamically display user-specific information in a clean and organized way.

CHAPTER: 5

BACK END OF SYSTEM

5.1 About ExpressJS

5.2 Why Use ExpressJS

5.3 About Back End

5.3.1: about MongoDB

5.3.2: How MongoDB Works

5.3.3: MongoDB Features

5.4 Why use MongoDB

5.1 About ExpressJS

ExpressJS is a fast, unopinionated, and minimalist web application framework for Node.js. It is widely used for building web applications and APIs (Application Programming Interfaces).

ExpressJS simplifies the process of managing HTTP requests, routes, middleware, and server-side logic, making backend development more efficient and streamlined.

Key Features of ExpressJS

- **Middleware Support:** Allows adding custom logic (like authentication, logging, etc.) during request processing.
- **Routing:** Provides simple and intuitive routing methods to handle GET, POST, PUT, DELETE requests.
- **Template Engines:** Supports various template engines like EJS, Pug, Handlebars, etc., for server-side rendering.
- **RESTful API Support:** Enables easy development of RESTful APIs for web and mobile applications.
- **Lightweight Framework:** It is lightweight and fast, allowing for quick development of web applications.
- **Error Handling:** Offers built-in error-handling middleware to manage runtime errors.

5.2 Why Use ExpressJS

- **Fast and Easy Development:** ExpressJS minimizes the complexity of writing backend code and accelerates the development process.
- **Minimal Setup:** Setting up a web server with ExpressJS requires minimal code and configuration.
- **Flexible and Scalable:** It is highly flexible and can be used for small to large-scale applications.
- **Open Source and Community Support:** Being open-source, it has strong community support and a vast range of third-party libraries.
- **Full-Stack Compatibility:** Works seamlessly with MongoDB, ReactJS, Node.js, and other technologies.

5.3 About Back End

The back end of a system refers to the server-side of an application where business logic, database management, authentication, and server responses are handled. It typically involves:

- **Database Management:** Managing and storing data in a database.
- **API Development:** Creating endpoints for the frontend to communicate with the server.
- **User Authentication:** Managing user sign-in, sign-out, and authorization.
- **Server Logic:** Handling the core business logic and processes.

5.3.1 About MongoDB

MongoDB is a widely used NoSQL (Non-Structured Query Language) database that stores data in flexible, JSON-like documents. It is designed for scalability, high availability, and ease of development.

Key Features of MongoDB

- **Document-Oriented:** Stores data in JSON-like documents with dynamic schemas.
- **Scalability:** Can handle large volumes of data with horizontal scalability.
- **Flexible Schema:** Allows dynamic changes in data structure without downtime.
- **High Performance:** Provides high read and write performance.
- **Indexing:** Allows indexing to enhance query performance.
- **Aggregation Framework:** Offers powerful data aggregation and analytics capabilities.

5.3.2 How MongoDB Works

MongoDB works by storing data in flexible, JSON-like documents called **BSON (Binary JSON)**. These documents are grouped into collections, similar to tables in relational databases. MongoDB does not require a fixed schema, allowing developers to modify data structures without affecting existing data.

- **Insertion:** Data is inserted in BSON format, allowing for dynamic and nested structures.
- **Querying:** MongoDB uses a powerful query language to retrieve data based on key-

value pairs.

- **Indexing:** Indexes are created on fields to optimize search operations.
- **Replication:** MongoDB supports replication, allowing data to be synchronized across multiple servers for high availability.
- **Sharding:** Large datasets are divided into smaller pieces (shards) to enhance performance.

5.3.3 MongoDB Features

- **Flexible Data Model:** No fixed schema allows easy modification of data structures.
- **Scalability:** Horizontal scaling ensures high performance with large datasets.
- **High Availability:** Replication ensures data is always available, reducing downtime.
- **Indexing:** Indexes improve the performance of read operations.
- **Document-Oriented:** Data is stored in documents rather than rows and columns.
- **Open-Source:** MongoDB is free to use and has a large active community.

5.4 Why Use MongoDB

- **Flexibility:** MongoDB's flexible schema design allows storing a variety of data types without a fixed structure.
- **Scalable Architecture:** Handles large-scale applications with distributed data and high traffic.
- **High-Speed Performance:** Ensures faster data retrieval and storage with document-based data models.
- **Real-time Data Processing:** Ideal for applications requiring real-time data processing, such as e-commerce, IoT, and analytics.
- **Open Source:** It is free and has a large community that provides continuous support and improvements.

The backend of the system plays a critical role in managing data, processing requests, and ensuring secure and efficient communication between the frontend and the database. Using **ExpressJS** and **MongoDB** together provides a powerful combination for developing fast, scalable, and efficient web applications.

CHAPTER 6

SYSTEM ANALYSIS AND SYSTEM DESIGN

6.1 System Analysis and Design

6.2 Systems Development Life Cycle (SDLC)

6.2.1: Needs of SDLC

6.2.2: SDLC Cycle

6.3 Spiral Model

6.4 Class Diagram

6.1 System Analysis and Design

System Analysis and Design (SAD) is the backbone of software development, guiding the creation of efficient and effective systems to address organizational needs. At its core, SAD is about understanding the current state of affairs within an organization, envisioning a better future, and architecting the pathway to get there.

During the analysis phase, SAD professionals delve deep into the existing systems and processes, meticulously identifying strengths, weaknesses, and opportunities for improvement. Through interviews, surveys, and observation, they gather requirements from stakeholders, ensuring that every voice is heard and every need is accounted for.

With requirements in hand, the design phase begins, where SAD experts transform abstract ideas into concrete plans. They define the system architecture, user interfaces, and data structures, ensuring that the design aligns closely with the gathered requirements and the overarching goals of the organization.

Implementation brings the design to life, as developers translate the blueprints into functional code. Rigorous testing follows, with each component scrutinized to ensure its reliability, security, and performance. User acceptance testing ensures that the system not only meets technical specifications but also satisfies the needs and expectations of its intended users.

Deployment marks the culmination of the SAD process, as the new system is introduced into the organization's ecosystem. With proper training and support, users adapt to the new system, embracing its capabilities and maximizing its potential to drive efficiency and innovation.

But the journey doesn't end there. Maintenance and support are ongoing responsibilities, as SAD professionals monitor the system, address issues, and incorporate updates to keep it running smoothly and meeting the evolving needs of the organization.

Throughout every phase of System Analysis and Design, collaboration, communication, and a relentless focus on the end-users are paramount. By aligning technology with business objectives and user needs, SAD professionals play a vital role in shaping the future success of organizations across industries.

6.2 Systems Development Life Cycle (SDLC)

A software life cycle model (also termed process model) is a pictorial and diagrammatic representation of the software life cycle. A life cycle model represents all the methods required to make a software product transit through its life cycle stages. It also captures the structure in which these methods are to be undertaken.

In other words, a life cycle model maps the various activities performed on a software product from its inception to retirement. Different life cycle models may plan the necessary development activities to phases in different ways. Thus, no element which life cycle model is followed, the essential activities are contained in all life cycle models though the action may be carried out in distinct orders in different life cycle models. During any life cycle stage, more than one activity may also be carried out.

6.2.1 Needs of SDLC

The development team must determine a suitable life cycle model for a particular plan and then observe to it.

Without using an exact life cycle model, the development of a software product would not be in a systematic and disciplined manner. When a team is developing a software product, there must be a clear understanding among team representative about when and what to do. Otherwise, it would point to chaos and project failure. This problem can be defined by using an example. Suppose a software development issue is divided into various parts and the parts are assigned to the team members. From then on, suppose the team representative is allowed the freedom to develop the roles assigned to them in whatever way they like. It is possible that one representative might start writing the code for his part, another might choose to prepare the test documents first, and some other engineer might begin with the design phase of the roles assigned to him. This would be one of the perfect methods for project failure.

A software life cycle model describes entry and exit criteria for each phase. A phase can begin only if its stage-entry criteria have been fulfilled. So without a

software life cycle model, the entry and exit criteria for a stage cannot be recognized. Without software life cycle models, it becomes tough for software project managers to monitor the progress of the project.

6.2.2 SDLC Cycle

SDLC Cycle represents the process of developing software. SDLC framework includes the following steps:

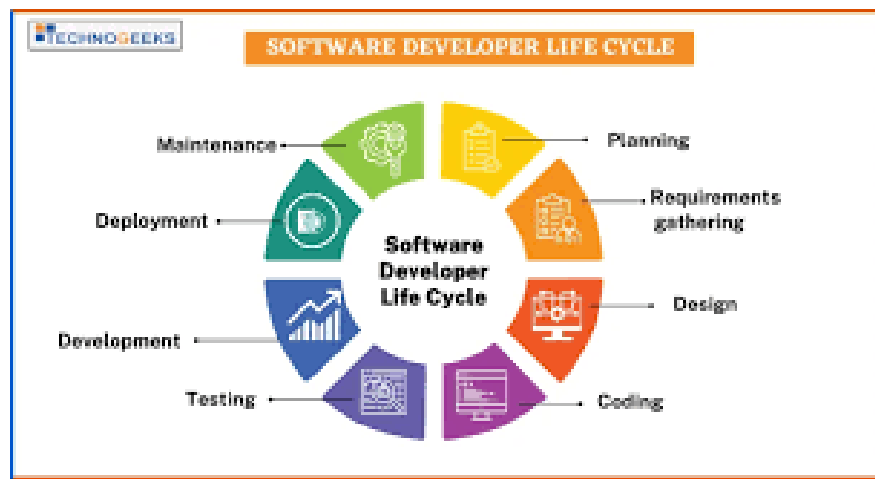


Fig 6.2.1: SDCL Cycle

- The stages of SDLC are as follows:

Stage1: Planning and requirement analysis

- Requirement Analysis is the most important and necessary stage in SDLC.
- The senior members of the team perform it with inputs from all the stakeholders and domain experts or SMEs in the industry.
- Planning for the quality assurance requirements and identifications of the risks associated with the projects is also done at this stage.

Stage2: Defining Requirements

- Once the requirement analysis is done, the next stage is to certainly represent and document the software requirements and get them accepted from the project stakeholders.
- This is accomplished through "SRS"- Software Requirement Specification document

which contains all the product requirements to be constructed and developed during the project life cycle.

Stage3: Designing the Software

- The next phase is about to bring down all the knowledge of requirements, analysis, and design of the software project. This phase is the product of the last two, like inputs from the customer and requirement gathering.

Stage4: Developing the project

- In this phase of SDLC, the actual development begins, and the programming is built. The implementation of design begins concerning writing code.
- Developers have to follow the coding guidelines described by their management and programming tools like compilers, interpreters, debuggers, etc. are used to develop and implement the code.

Stage5: Testing

- After the code is generated, it is tested against the requirements to make sure that the products are solving the needs addressed and gathered during the requirements stage.
- During this stage, unit testing, integration testing, system testing, acceptance testing are done.

Stage6: Deployment

- Once the software is certified, and no bugs or errors are stated, then it is deployed.
- Then based on the assessment, the software may be released as it is or with suggested enhancement in the object segment. After the software is deployed, then its maintenance begins.

Stage7: Maintenance

- Once when the client starts using the developed systems, then the real issues come up and requirements to be solved from time to time. This procedure where the care is taken for the developed product is known as maintenance.

6.3 Spiral Model

The Spiral Model is a Software Development Life Cycle (SDLC) model that provides a systematic and iterative approach to software development. In its diagrammatic representation, looks like a spiral with many loops. The exact number of loops of the spiral is unknown and can vary from project to project. Each loop of the spiral is called a Phase of the software development process.

The exact number of phases needed to develop the product can be varied by the project manager depending upon the project risks.

As the project manager dynamically determines the number of phases, the project manager has an important role in developing a product using the spiral model.

It is based on the idea of a spiral, with each iteration of the spiral representing a complete software development cycle, from requirements gathering and analysis to design, implementation, testing, and maintenance.

The Spiral Model is a risk-driven model, meaning that the focus is on managing risk through multiple iterations of the software development process. **It consists of the following phases:**

- **Planning**

- The first phase of the Spiral Model is the planning phase, where the scope of the project is determined and a plan is created for the next iteration of the spiral.

- **Risk Analysis**

- In the risk analysis phase, the risks associated with the project are identified and evaluated.

- **Engineering**

- In the engineering phase, the software is developed based on the requirements gathered in the previous iteration.

- **Evaluation**

- In the evaluation phase, the software is evaluated to determine if it

meets the customer's requirements and if it is of high quality.

- **Planning**

- The next iteration of the spiral begins with a new planning phase, based on the results of the evaluation.

The Spiral Model is often used for complex and large software development projects, as it allows for a more flexible and adaptable approach to software development. It is also well-suited to projects with significant uncertainty or high levels of risk.

The Radius of the spiral at any point represents the expenses(cost) of the project so far, and the angular dimension represents the progress made so far in the current phase.

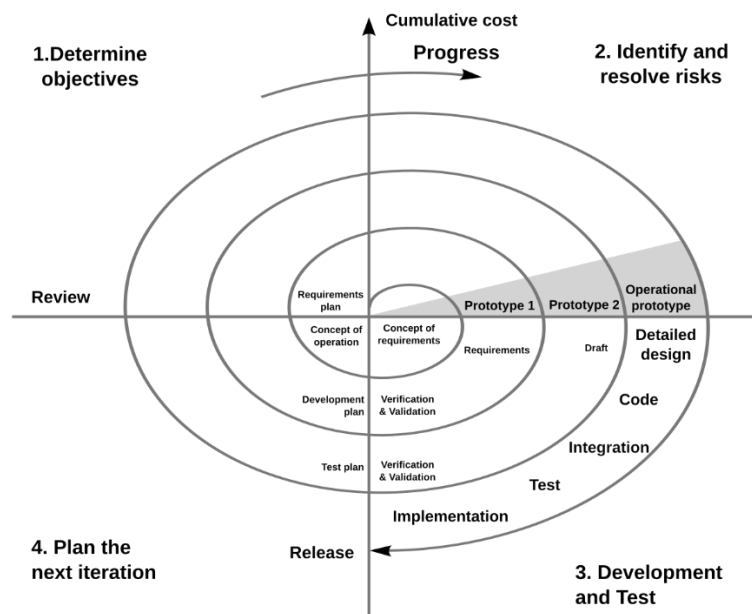


Fig. 6.3.1: Spiral Model

Each phase of the Spiral Model is divided into four quadrants as shown in the above figure. The functions of these four quadrants are discussed below:

- **Objectives determination and identify alternative solutions:**

- Requirements are gathered from the customers and the objectives are identified, elaborated, and analyzed at the start of every phase. Then alternative solutions possible for the phase are proposed in this quadrant.

- **Identify and resolve Risks:**

- During the second quadrant, all the possible solutions are evaluated to select the best possible solution. Then the risks associated with that solution are identified and the risks are resolved using the best possible strategy. At the end of this quadrant, the Prototype is built for the best possible solution.

- **Develop the next version of the Product:**

- During the third quadrant, the identified features are developed and verified through testing. At the end of the third quadrant, the next version of the software is available.

- **Review and plan for the next Phase:**

- In the fourth quadrant, the Customers evaluate the so-far developed version of the software. In the end, planning for the next phase is started.

6.4 Class Diagram

The class diagram depicts a static view of an application. It represents the types of objects residing in the system and the relationships between them. A class consists of its objects, and also it may inherit from other classes. A class diagram is used to visualize, describe, document various different aspects of the system, and also construct executable software code.

It shows the attributes, classes, functions, and relationships to give an overview of the software system. It constitutes class names, attributes, and functions in a separate compartment that helps in software development. Since it is a collection of classes, interfaces, associations, collaborations, and constraints, it is termed as a structural diagram.

- **Purpose of Class Diagrams**

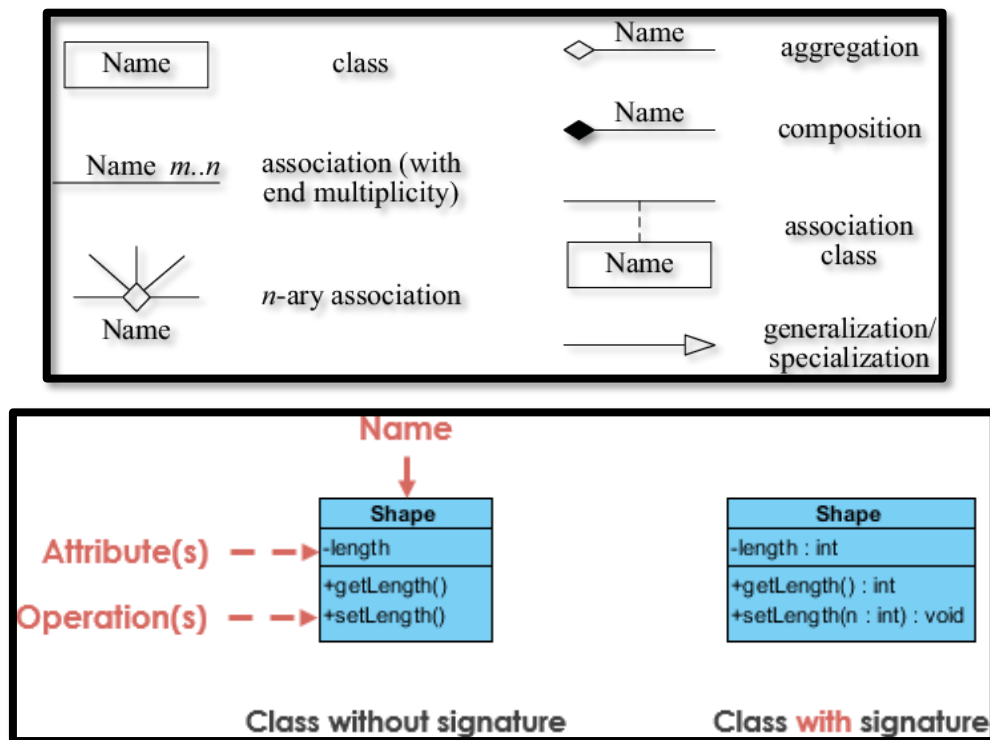
The main purpose of class diagrams is to build a static view of an application. It is the only diagram that is widely used for construction, and it can be mapped with object-oriented languages. It is one of the most popular UML diagrams. Following are the

purpose of class diagrams given below:

- It analyses and designs a static view of an application.
- It describes the major responsibilities of a system.
- It is a base for component and deployment diagrams.
- It incorporates forward and reverse engineering.

• Benefits of Class Diagrams

- It can represent the object model for complex systems.
- It reduces the maintenance time by providing an overview of how an application is structured before coding.
- It provides a general schematic of an application for better understanding.
- It represents a detailed chart by highlighting the desired code, which is to be programmed.
- It is helpful for the stakeholders and the developers.
- Symbols used in Class:



• Fig. 6.4.1 Symbols Used in Class

CHAPTER: 7

DATA DICTIONARY

7.1 Introduction

7.2 List of Tables

7.1.1: User_Collection

7.1.2: Gym_Collection

7.1.3: Booking_Collection

7.1.4: Otp_Collection

7.1.5: Session_Collection

7.1 Introduction

Data Dictionary is an important part of a project or system which contains all definition of elements in the system. In Data Dictionary you will find a list of all elements composing the data flowing through a system.

- The major elements of a system:
 - Data Flow
 - Data Store
 - Processes
- The data dictionary stores all details and description of these elements.
- The data dictionary provides additional information about the system.
- The data dictionary contains the data about data e.g. Student is a data and this student belongs to this course is a description of STUDENT data which is stored in data dictionary.
- Why is Data Dictionary important?
 - To manage the details in large system.
 - To communicate a common meaning for all system elements.
 - To document the features of system.
 - To determine where system changes should made.
 - To locate errors and omissions in the system.

7.2 List of Tables:

Table 7.2.1: User_Collection

- **Description: This table stores data related to the user module of the gym recommended system.**

[Table 7.2.1: User_Collection]

Field Name	Data Type	Description
_id	ObjectId	Unique identifier for each user
name	String	Full name of the user
email	String	User's email address (used for login)
password	String (Hashed)	Encrypted password for user authentication
role	String	Defines user role (User, Trainer, Admin)

preferences	Array of String	User's fitness preferences like Yoga, Cardio, etc.
createdAt	Date	Account creation date
updatedAt	Date	Last profile update date

Table 7.2.2: Gym_Collection

- **Description: This table stores data related to the gym module of the gym recommended system.**

[Table 7.2.2: Gym_Collection]

Field Name	Data Type	Description
_id	ObjectId	Unique identifier for each gym
name	String	Name of the gym
location	String	Address or locality of the gym
facilities	Array of String	Available facilities
trainers	Array of ObjectId	Reference to trainers working at the gym
subscriptionFee	Number	Monthly fee or flexible plan fee
contactNumber	String	Contact number for inquiries
ratings	Number	Average rating of the gym
createdAt	Date	Date when the gym was added
updatedAt	Date	Last update of gym details

Table 7.2.3: Booking_Collection

- **Description: This table stores data related to the booking gym module of the gym recommended system.**

[Table 7.2.3: Booking_Collection]

Field Name	Data Type	Description
_id	ObjectId	Unique identifier for each booking
userId	ObjectId	Reference to the user who made the booking
gymId	ObjectId	Reference to the gym being booked
bookingDate	Date	Date of booking
duration	String	Booking duration (1 week, 1 month, etc.)
paymentStatus	String	Status of payment (Paid, Pending, Failed)

Table 7.2.4: Otp_Collection

- **Description:** This table stores data related to the otp which are used in authentication process of the gym recommended system.

[Table 7.2.4: Otp_Collection]

Field Name	Data Type	Description
_id	ObjectId	Unique identifier for each gym
email	String	Email of the user
otp	Number	6 digit unique one time password

Table 7.2.5: Session_Collection

- **Description:** This table stores data related to the session or token which are used in authentication process of the gym recommended system.

[Table 7.2.5: Session_Collection]

Field Name	Data Type	Description
_id	ObjectId	Unique identifier for each session.
expires	Date	Expiration date and time of the session.
session	Object	Contains session data as an object.
session.cookie	Object	Contains cookie-related information of the session.
session.accessToken	String (JWT)	JWT token provided after successful login.

CHAPTER:8

TESTING

8.1 Testing Plan

8.2 Testing Strategies

8.3 Testing Method

8.4 Test Case

8.1 Testing Plan

Great! Let's outline a detailed testing plan. Since you haven't specified the exact product or system you're testing, I'll provide a generic outline that you can adapt to your specific needs. Here's a structured testing plan:

- **Introduction:**

- Provide an overview of the testing plan.
- Include the objectives, scope, and key stakeholders.

- **Purpose:**

- Define the purpose of testing.
- Clarify what aspects of the product/system will be tested.

- **Scope:**

- Describe the boundaries of the testing.
- Identify the features, functionalities, or components included in the scope.
- Mention any excluded items or functionalities.

- **Testing Goals:**

- List the specific goals or objectives of the testing.
- These goals should align with the overall project objectives and stakeholder requirements.

- **Testing Strategy:**

- Outline the overall testing approach.
- Specify the types of testing to be conducted (e.g., functional testing, usability testing, performance testing, security testing).
- Describe any testing methodologies or frameworks to be followed (e.g., Agile, Waterfall, DevOps).

- **Test Deliverables:**

- List the documents, reports, or artifacts that will be produced during testing.

- Include items such as test plans, test cases, test scripts, and test reports.
- **Testing Environment:**
 - Specify the hardware, software, and network configurations required for testing.
 - Detail any test environment setup or configurations needed.
- **Test Cases:**
 - Develop detailed test cases for each identified test scenario.
 - Include preconditions, steps to execute, expected results, and post-conditions.
 - Organize test cases by priority and complexity.
- **Test Execution:**
 - Describe how test cases will be executed.
 - Assign responsibilities for test execution.
 - Define the criteria for determining when testing is complete.
- **Defect Management:**
 - Explain how defects/issues will be logged, tracked, and managed.
 - Define severity and priority levels for defects.
 - Outline the defect resolution process.
- **Risks and Mitigation:**
 - Identify potential risks associated with testing.
 - Propose mitigation strategies for each identified risk.
- **Schedule:**
 - Create a testing timeline or schedule.
 - Allocate time for test preparation, execution, and reporting.
 - Consider any dependencies or constraints that may impact the schedule.
- **Roles and Responsibilities:**
 - Define the roles and responsibilities of team members involved in testing.

- Specify who is responsible for test planning, execution, documentation, and communication.
- **Communication Plan:**
 - Outline how communication will be managed throughout the testing process.
 - Identify key stakeholders and their communication preferences.
 - Specify the frequency and format of status updates and reports.
- **Review and Approval:**
 - Establish a process for reviewing and approving the testing plan.
 - Identify the stakeholders responsible for review and approval.
- **Conclusion:**
 - Summarize the key points of the testing plan.
 - Confirm alignment with project goals and stakeholder expectations.
- **Appendices:**
 - Include any supplementary information or reference materials relevant to the testing plan.

Once you've drafted your testing plan, be sure to review it with your team and stakeholders to ensure clarity, alignment, and agreement before proceeding with testing activities.

8.2 Testing Strategies

Testing strategies are crucial for ensuring that the testing process is efficient, thorough, and effective. Here's a detailed breakdown of various testing strategies:

- **Unit Testing:**

This strategy involves testing individual components or units of code in isolation.

- **Focus:** It verifies the correctness of each unit's behavior and functionality.
- **Approach:** Developers typically write unit tests using testing frameworks like JUnit (for Java), NUnit (for .NET), or Jest (for NodeJs).
- **Benefits:** Unit testing helps identify defects early in the development cycle

and promotes code modularity and reusability.

- **Integration Testing:**

Integration testing verifies the interactions between different units or modules within the system.

- Focus: It ensures that integrated components work together as expected.
- Approach: Integration tests can be conducted using techniques such as top- down, bottom-up, or a combination of both.
- Benefits: Integration testing detects interface defects and integration issues early, reducing the risk of major failures during system testing.

- **System Testing:**

System testing evaluates the entire system as a whole, including all integrated components.

- Focus: It validates the system against its functional and non-functional requirements.
- Approach: System testing can include functional testing, usability testing, performance testing, security testing, and more.
- Benefits: System testing ensures that the software meets user expectations and performs reliably in its intended environment.

- **Acceptance Testing:**

Acceptance testing determines whether the system meets the acceptance criteria defined by stakeholders.

- Focus: It validates the system from the end-user's perspective and assesses its readiness for deployment.
- Approach: Acceptance testing can be performed using techniques such as alpha testing, beta testing, or user acceptance testing (UAT).
- Benefits: Acceptance testing ensures that the software satisfies business requirements and aligns with user needs.

- **Regression Testing:**

Regression testing verifies that recent code changes have not adversely affected existing functionality.

- Focus: It ensures that previously tested features still work correctly after modifications or enhancements.
- Approach: Automated regression testing tools can help efficiently re-

run existing test cases to detect regressions.

Benefits: Regression testing helps maintain software quality and prevents the introduction of unintended defects during development.

- **Risk-Based Testing:**

Risk-based testing prioritizes testing efforts based on the likelihood and impact of potential failures.

- Focus: It focuses testing resources on high-risk areas of the system.
- Approach: Risk analysis techniques, such as Failure Mode and Effects Analysis (FMEA), help identify critical areas for testing.
- Benefits: Risk-based testing optimizes testing efforts by directing resources where they are most needed, reducing the likelihood of high-impact failures.

- **Security Testing:**

Security testing identifies vulnerabilities and weaknesses in the system's security controls.

- Focus: It evaluates the system's ability to protect data, prevent unauthorized access, and resist attacks.
- Approach: Security testing includes techniques such as penetration testing, vulnerability scanning, and code reviews.
- Benefits: Security testing helps mitigate security risks, safeguard sensitive information, and maintain the integrity of the system.

- **Usability Testing:**

Usability testing assesses the system's user interface and user experience (UI/UX).

- Focus: It evaluates how easily users can interact with the system and accomplish their tasks.
- Approach: Usability testing involves observing real users as they interact with the system and collecting feedback through surveys or interviews.

- Benefits: Usability testing identifies usability issues, improves user satisfaction, and enhances the overall user experience.

Each of these testing strategies plays a critical role in ensuring the quality, reliability, and performance of software systems. The selection and combination of testing strategies depend on factors such as project requirements, constraints, and risks. It's essential to tailor the testing approach to the specific needs of the project and continuously adapt it throughout the software development lifecycle.

8.3 Testing Method

Testing methods refer to the specific approaches or techniques used to conduct testing activities. These methods help ensure thorough coverage, accuracy, and effectiveness in validating the software's functionality, performance, and other quality attributes. Here are some common testing methods:

- **Black Box Testing:**

In black box testing, the tester does not have access to the internal code or logic of the software. Instead, tests are designed based on the software's specifications and requirements.

- Approach: Test cases are created to validate inputs, outputs, and system behavior without knowledge of the internal implementation.
- Benefits: Black box testing focuses on the software's external behavior, allowing testers to identify issues from a user's perspective.

- **White Box Testing:**

White box testing, also known as glass box or clear box testing, involves examining the internal structure and logic of the software.

- Approach: Test cases are designed based on an understanding of the code, control flow, and data flow within the software.
- Benefits: White box testing provides insights into code coverage, control flow, and error handling, helping identify issues related.

- **Gray Box Testing:**

Gray box testing combines elements of both black box and white box testing. Testers have partial knowledge of the internal workings of the software.

- Approach: Test cases are designed based on a limited understanding of the internal code or structure, allowing testers to target specific areas of interest.
- Benefits: Gray box testing leverages both external and internal perspectives, providing a balance between test coverage and depth.

- **Manual Testing:**

Manual testing involves human testers executing test cases manually without the use of automation tools.

- Approach: Testers interact with the software interface, input data, and observe outputs to validate its behavior.
- Benefits: Manual testing allows for exploratory testing, subjective evaluation of user experience, and validation of complex scenarios that may be challenging to automate.

- **Automated Testing:**

Automated testing involves the use of software tools to execute pre-scripted tests, compare actual outcomes with expected results, and generate test reports.

- Approach: Test scripts are created to automate repetitive test scenarios, regression tests, and performance tests.
- Benefits: Automated testing improves efficiency, repeatability, and coverage of testing activities, especially for large-scale projects and continuous integration pipelines.

These testing methods can be combined and adapted based on project requirements, constraints, and goals. Effective testing often involves a combination of multiple methods to achieve comprehensive coverage and ensure software quality.

8.4 Test Case

A test case is a detailed set of conditions or steps that are designed to verify the functionality, behavior, or performance of a software application or system. Test cases are used to validate whether the software meets specific requirements and functions correctly under various scenarios. Here's a breakdown of what a test case typically includes:

- **Test Case ID:** A unique identifier for the test case, usually in the form of a combination of letters and numbers.
- **Test Case Title/Name:** A descriptive title or name that summarizes the purpose or objective of the test case.
- **Description:** A brief description or summary of what the test case is intended to achieve.
- **Preconditions:** Any prerequisites or conditions that must be met before executing the test case, such as system configurations, data setup, or user permissions.
- **Test Steps:** A sequence of step-by-step instructions outlining the actions to be performed to execute the test case. Each step should be clear, concise, and unambiguous.
- **Input Data:** The specific data values, parameters, or inputs that will be used during the test execution.
- **Expected Results:** The expected outcomes or behaviors that the system should exhibit when the test case is executed successfully. This includes both observable results and system responses.
- **Actual Results:** The actual outcomes observed during test execution. Testers record the results obtained after executing the test steps.
- **Pass/Fail Criteria:** Criteria used to determine whether the test case passes or fails based on a comparison between the actual results and the expected results.
- **Priority:** The priority level assigned to the test case, indicating its relative importance or urgency in the testing process.
- **Severity:** The severity level assigned to the test case, indicating the impact of a failure on the system or its users. Common severity levels include critical, major, minor, and trivial.

- **Test Environment:** Information about the test environment, including hardware, software, configurations, and dependencies required for test execution.
- **Test Data:** Any additional test data or conditions relevant to the test case, such as boundary values, edge cases, or negative scenarios.
- **Attachments/References:** Any supporting documents, screenshots, logs, or references that provide additional context or details for the test case.
- **Notes/Comments:** Any additional notes, comments, or observations relevant to the test case, such as insights, assumptions, or recommendations.

By documenting test cases in a structured and standardized format, teams can ensure consistent testing practices, facilitate test execution, and maintain comprehensive test cover throughout the software development lifecycle. Additionally, well-written test cases serve as valuable documentation for future reference, regression testing, and knowledge transfer within the team.

CHAPTER: 9

SNAPSHOT OF WEBSITE

9.1 Admin Snapshot

9.2 Owner Snapshot

9.3 Member Snapshot

9.1 Admin Snapshot

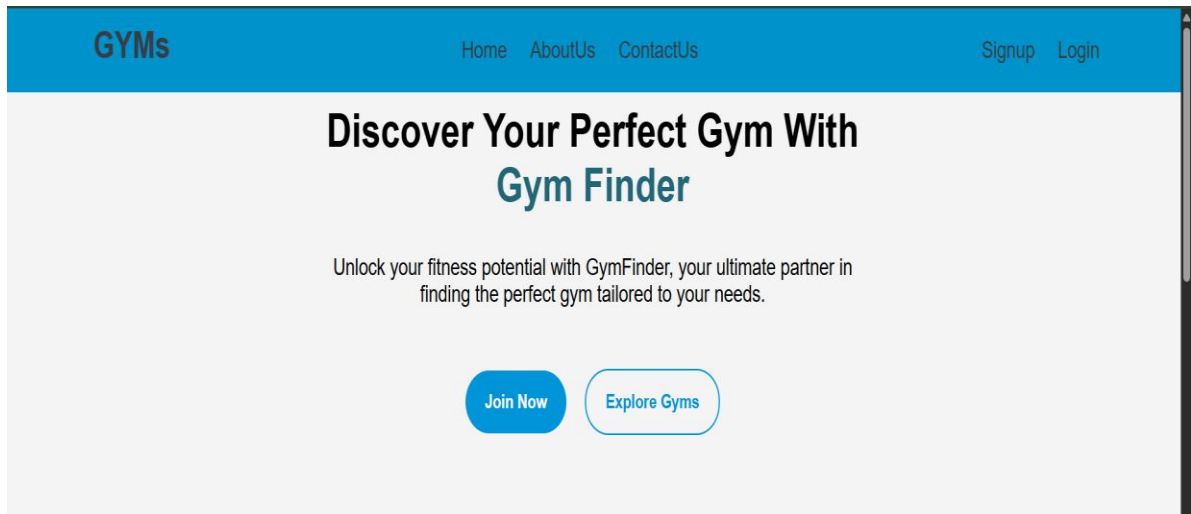


Fig. 9.1.1 Navbar and Hero Section

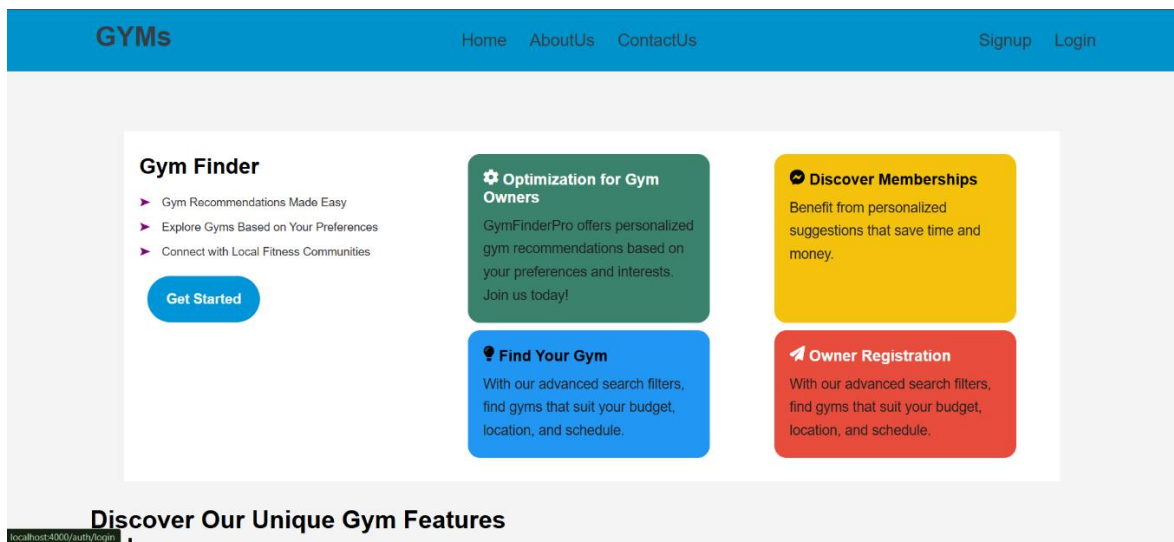


Fig. 9.1.2 Info Section

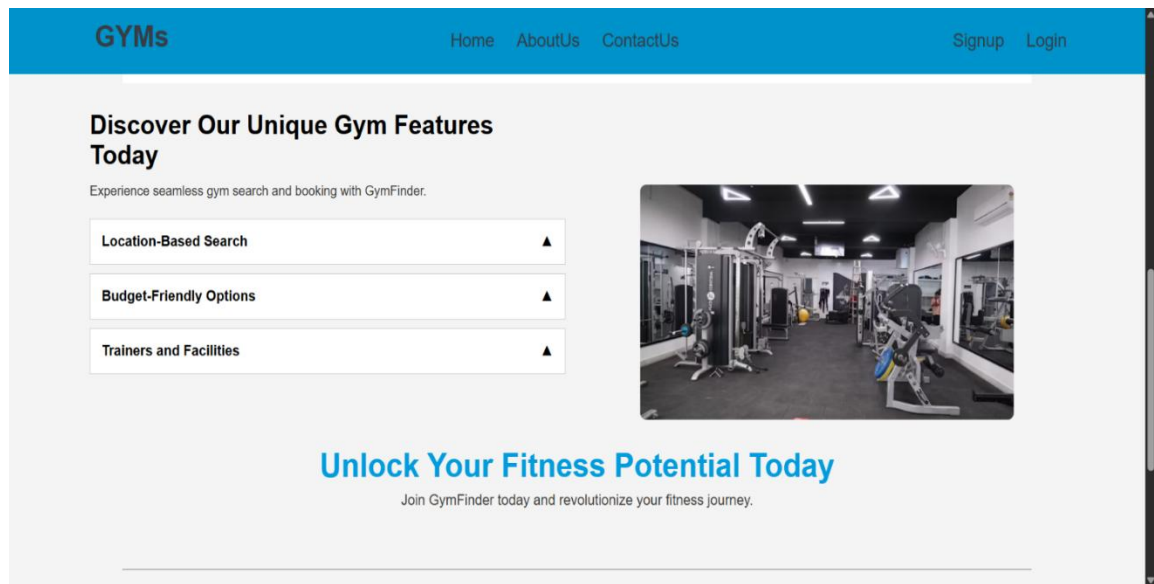


Fig. 9.1.3 Unique Feature

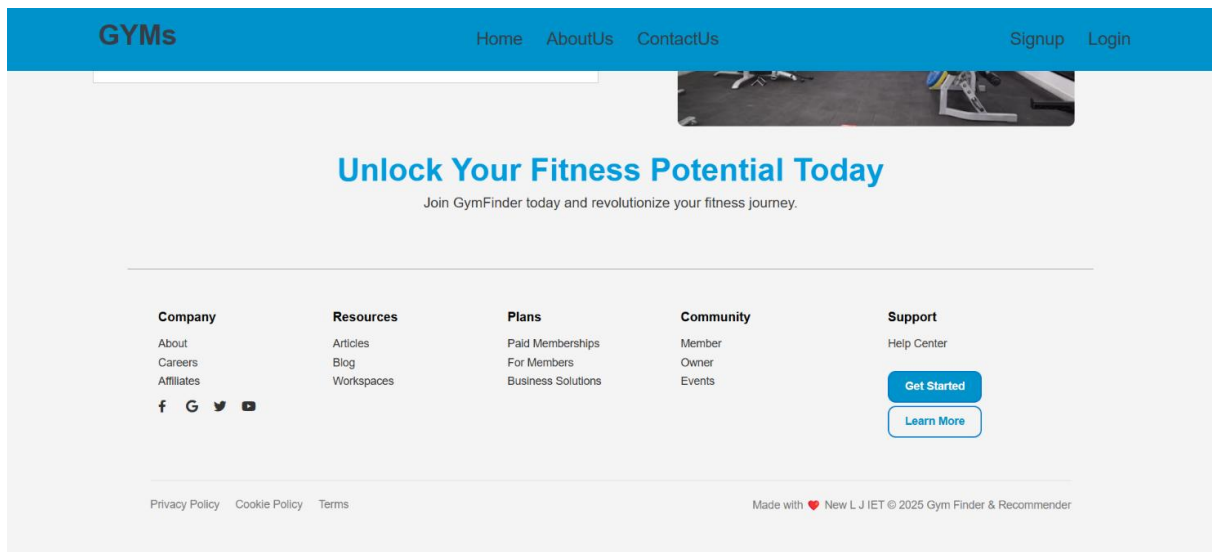


Fig. 9.1.4 Footer section

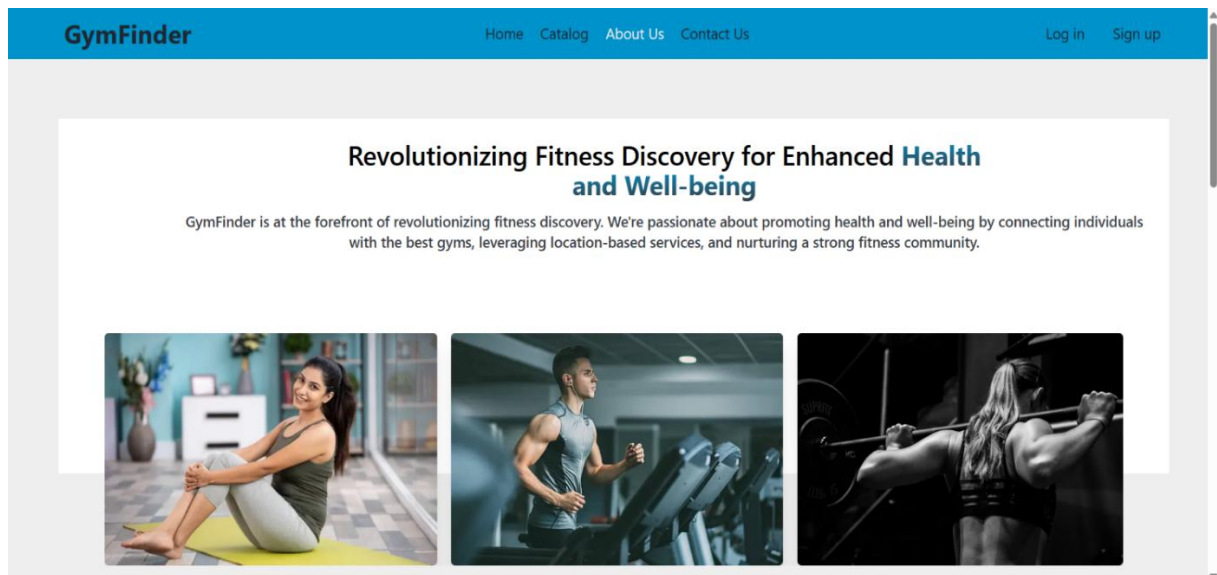


Fig. 9.1.5 About Hero section

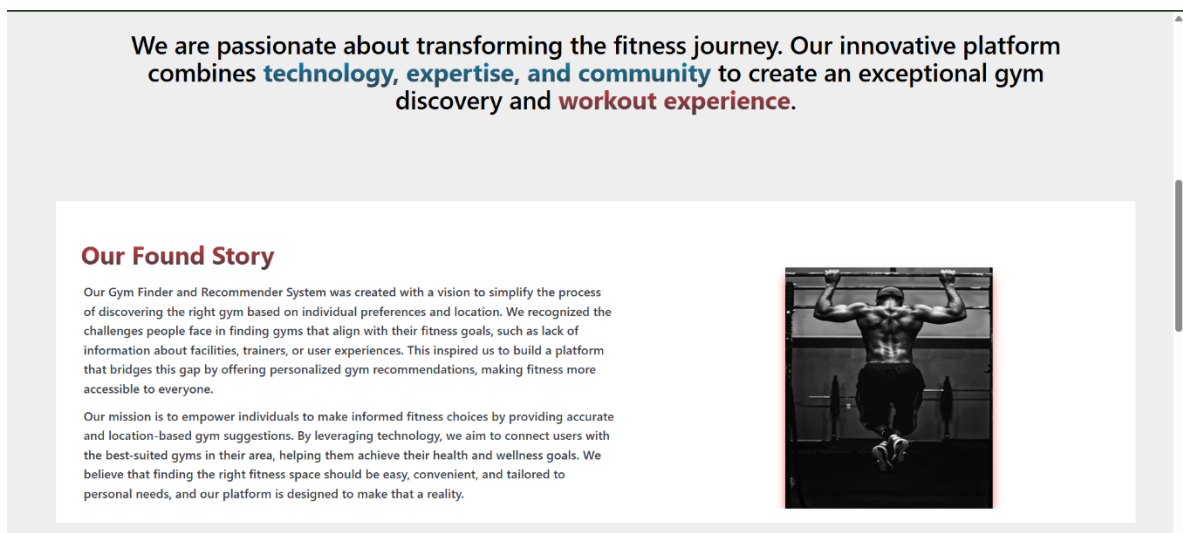
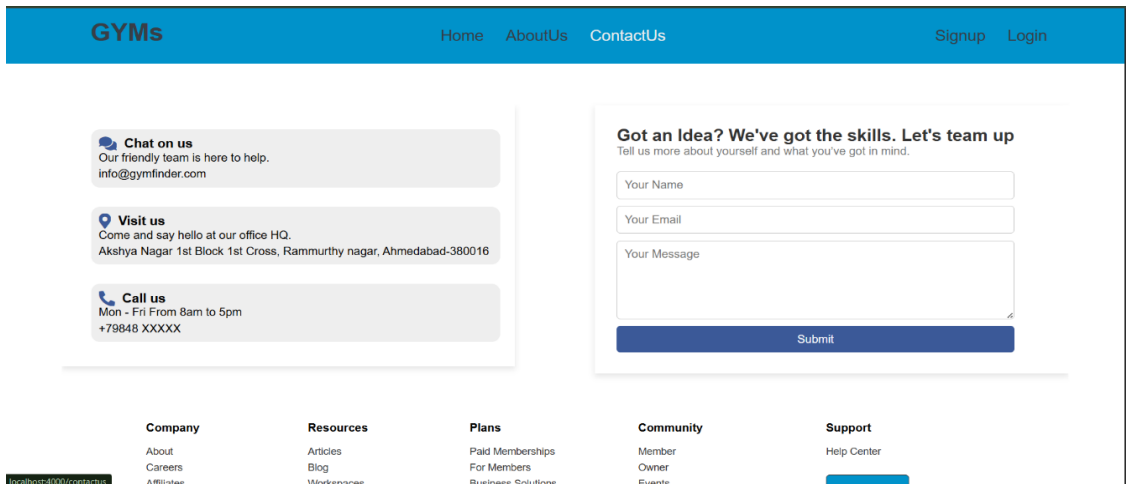


Fig. 9.1.6 Found Story section



GYMs Home AboutUs ContactUs Signup Login

Chat on us
Our friendly team is here to help.
info@gymfinder.com

Visit us
Come and say hello at our office HQ.
Akshya Nagar 1st Block 1st Cross, Rammurthy nagar, Ahmedabad-380016

Call us
Mon - Fri From 8am to 5pm
+79848 XXXXX

Got an Idea? We've got the skills. Let's team up
Tell us more about yourself and what you've got in mind.

Your Name

Your Email

Your Message

Submit

Company
About
Careers
Affiliates

Resources
Articles
Blog
Workspaces

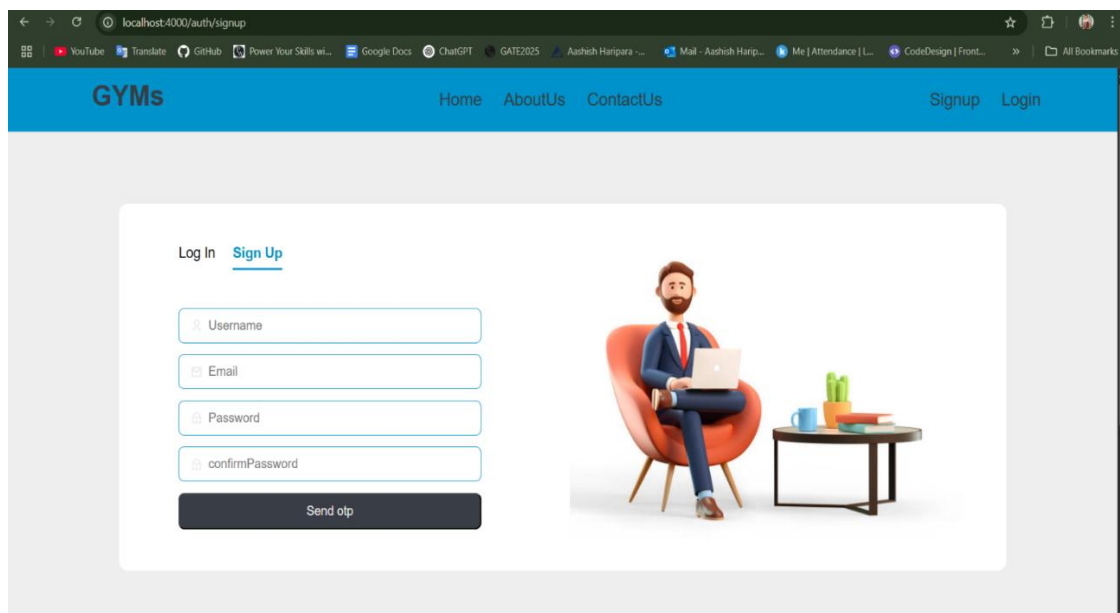
Plans
Paid Memberships
For Members
Business Solutions

Community
Member
Owner
Events

Support
Help Center
[Get Started](#)

localhost:4000/contactus

Fig. 9.1.7 Contact Us Page



localhost:4000/auth/signup

GYMs Home AboutUs ContactUs Signup Login

Log In Sign Up

Send otp

Illustration of a man sitting in a red armchair with a laptop, next to a small table with a cactus and books.

Fig. 9.1.8 Sign Up Page

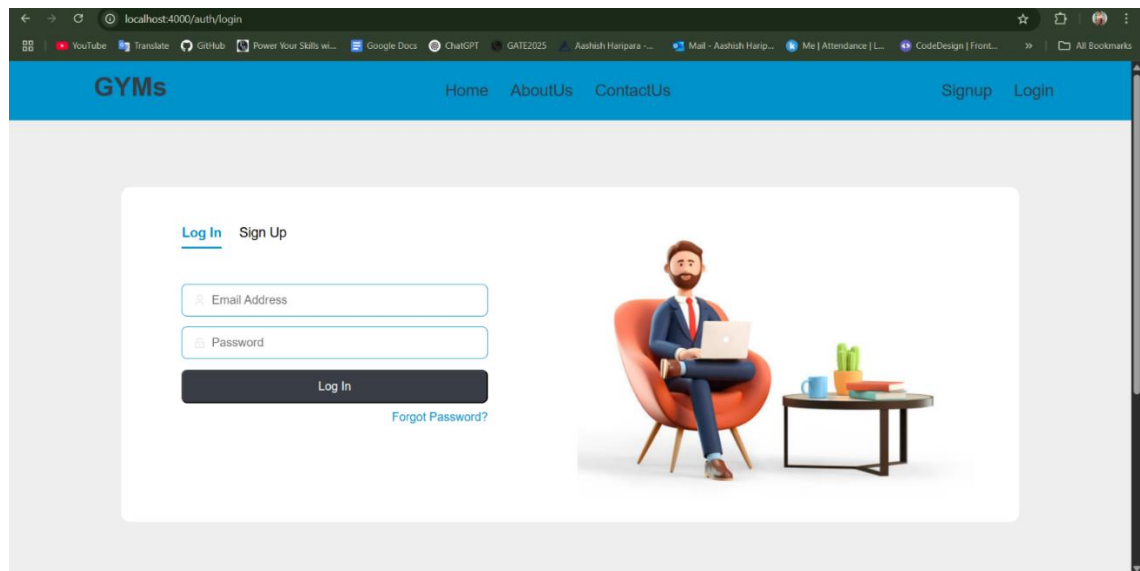


Fig. 9.1.9 Login Page

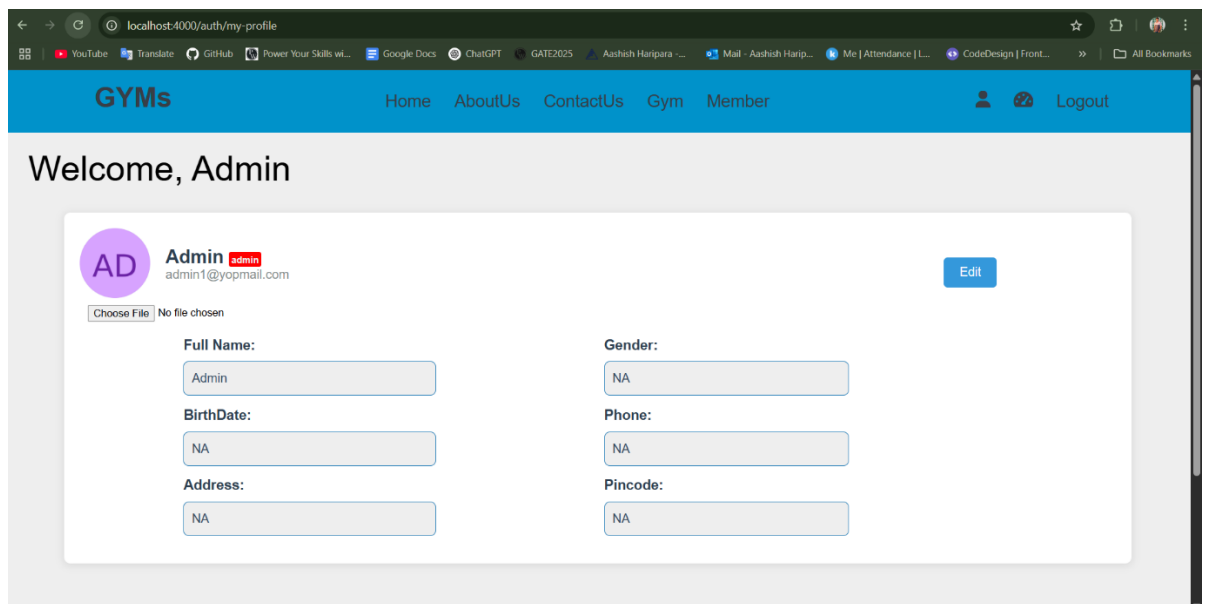


Fig. 9.2.1 Admin Profile Page

GYMs

Home AboutUs ContactUs Gym Member Logout

Welcome, Admin

1 Total Gyms

5 Total Members

Gym Name	Owner Name	Gym Capacity
One Gym	Sanjay	150

MEMBER ID	MEMBER NAME	MEMBER TYPE	MEMBERSHIP STATUS
O1	Kartik Vyas	owner	☒
M1	Manish	member	☒
M2	Sanjay	owner	☒
O2	Akshar	owner	☒

Fig. 9.2.2 Admin Dashboard Page

GYMs

Home AboutUs ContactUs Gym Member Logout

Gym Details

Search.. Submit Reset

Gym Name	Owner Name	Location	Pricing
One Gym	Sanjay	Ahmedabad	₹1200/month

Select Number of Data

Prev page 1 of 1 Next

Fig. 9.2.3 Admin Gym Details Page

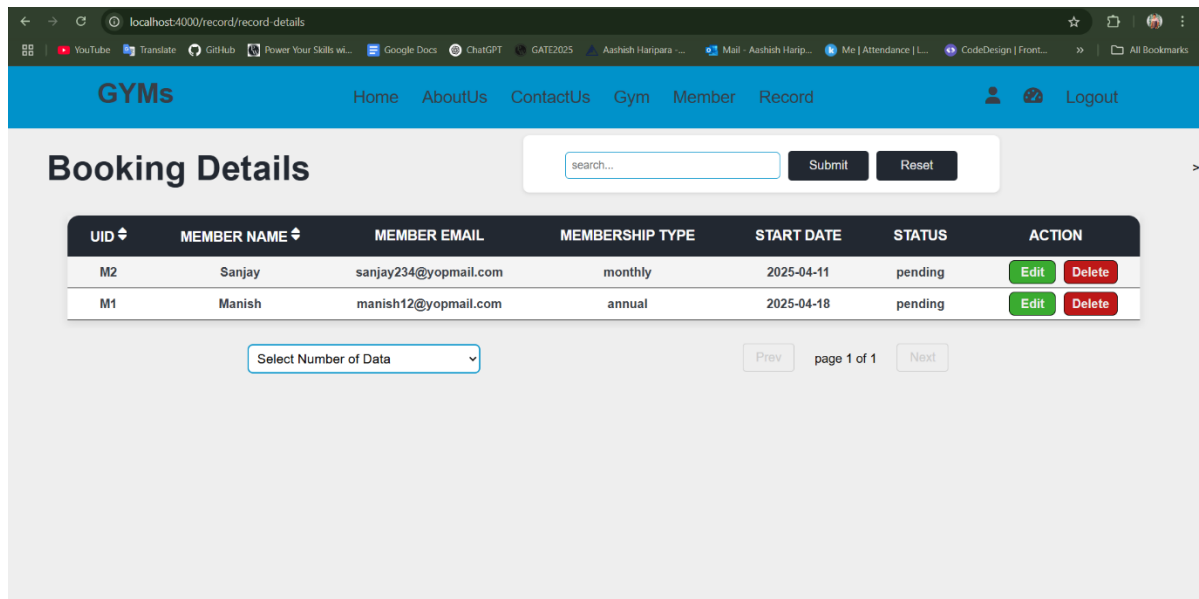


Fig. 9.2.4 Admin Member Details Page

9.3 Owner Snapshot:

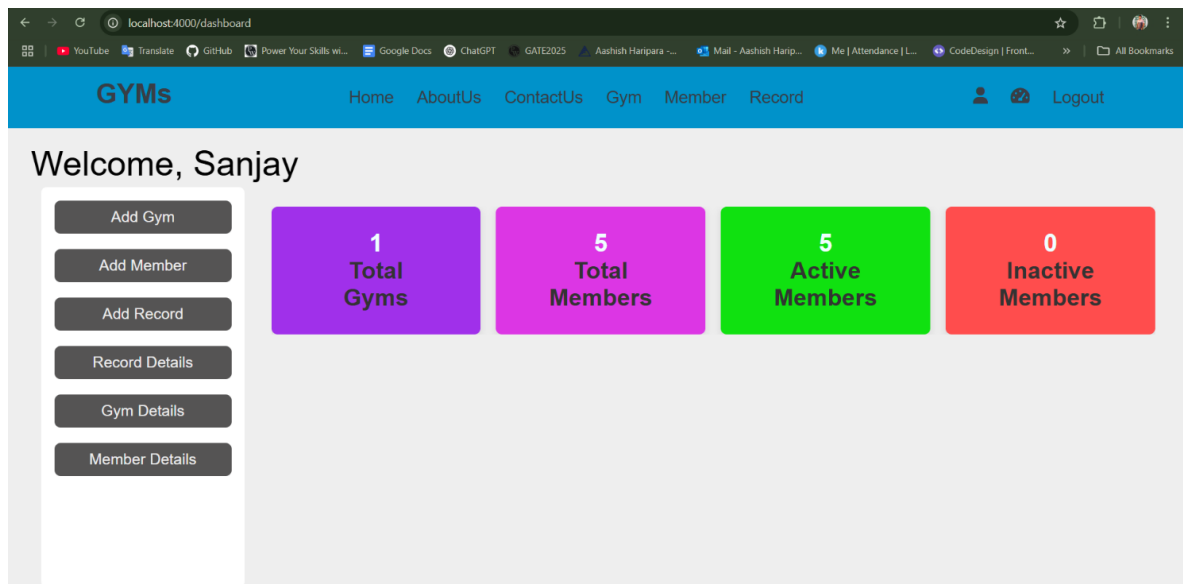


Fig. 9.3.1 Owner Dashboard Page

Fig. 9.3.2 Add member Page

UID	MEMBER NAME	MEMBER EMAIL	MEMBERSHIP TYPE	START DATE	STATUS	ACTION
M2	Sanjay	sanjay234@yopmail.com	monthly	2025-04-11	pending	<button>Edit</button> <button>Delete</button>
M1	Manish	manish12@yopmail.com	annual	2025-04-18	pending	<button>Edit</button> <button>Delete</button>

Fig. 9.3.3 Booking Record Page

9.4 Member Snapshot:

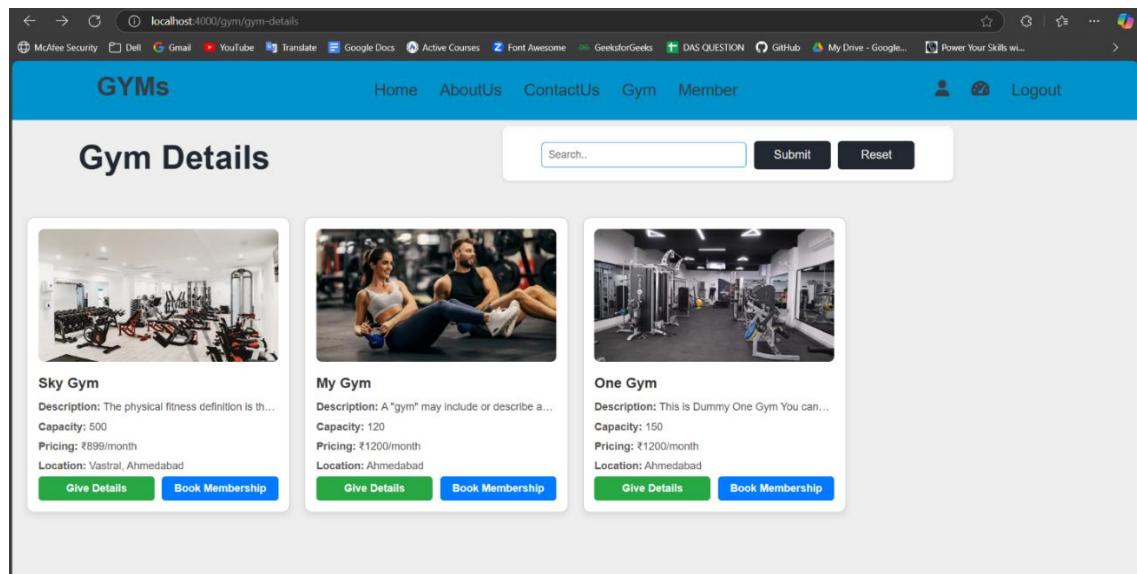


Fig. 9.4.1 Member Gym Details Page

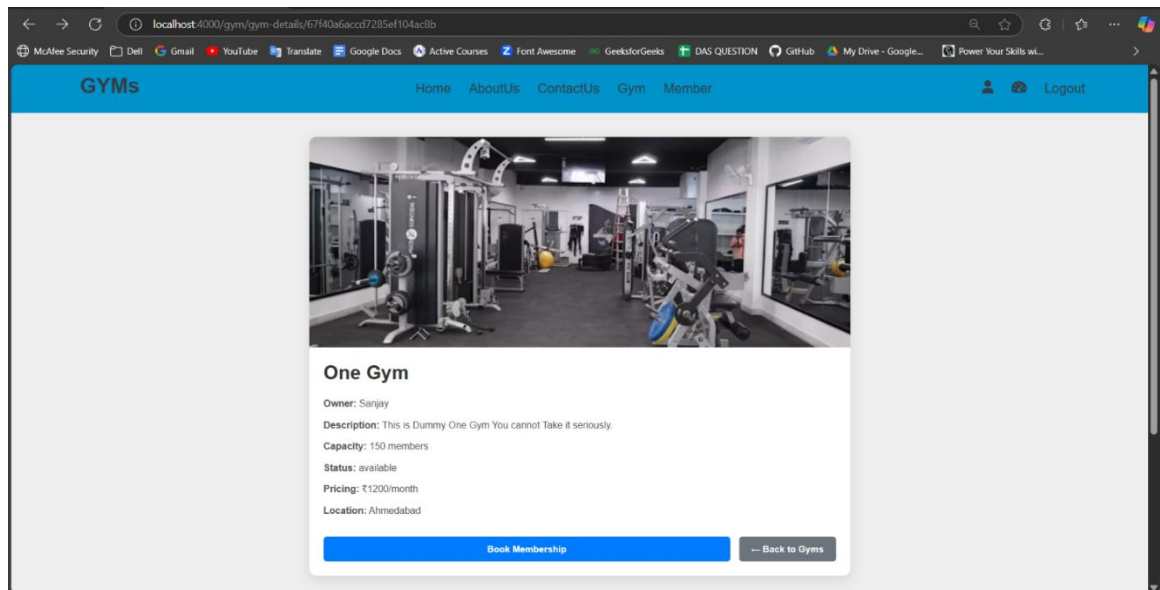


Fig. 9.4.2 One Gym Details Page

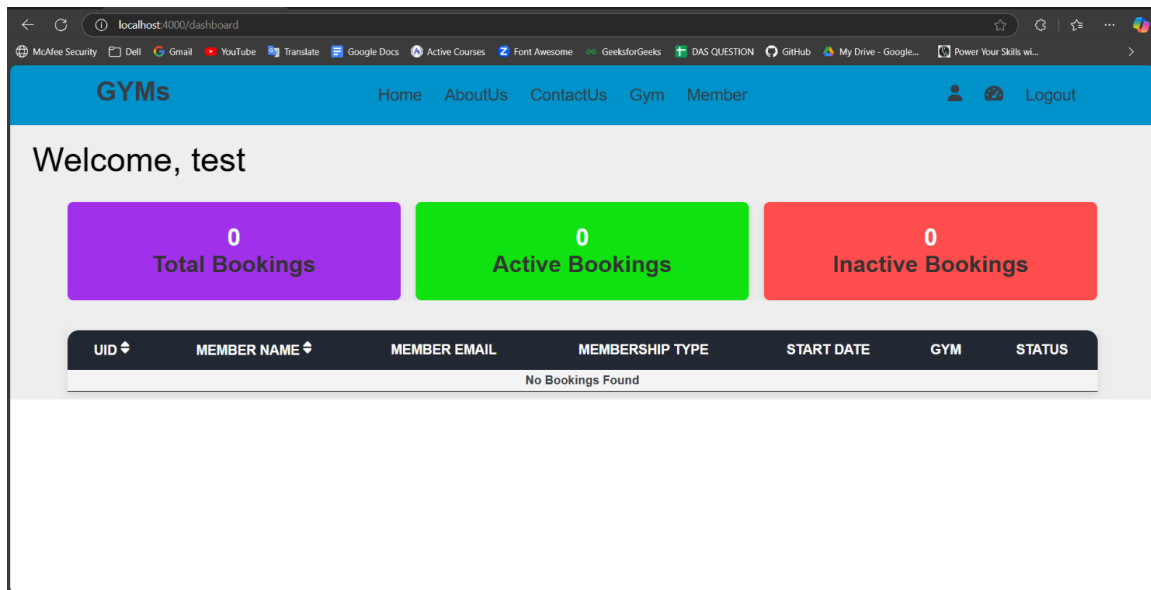


Fig. 9.4.3 Member Dashboard Page

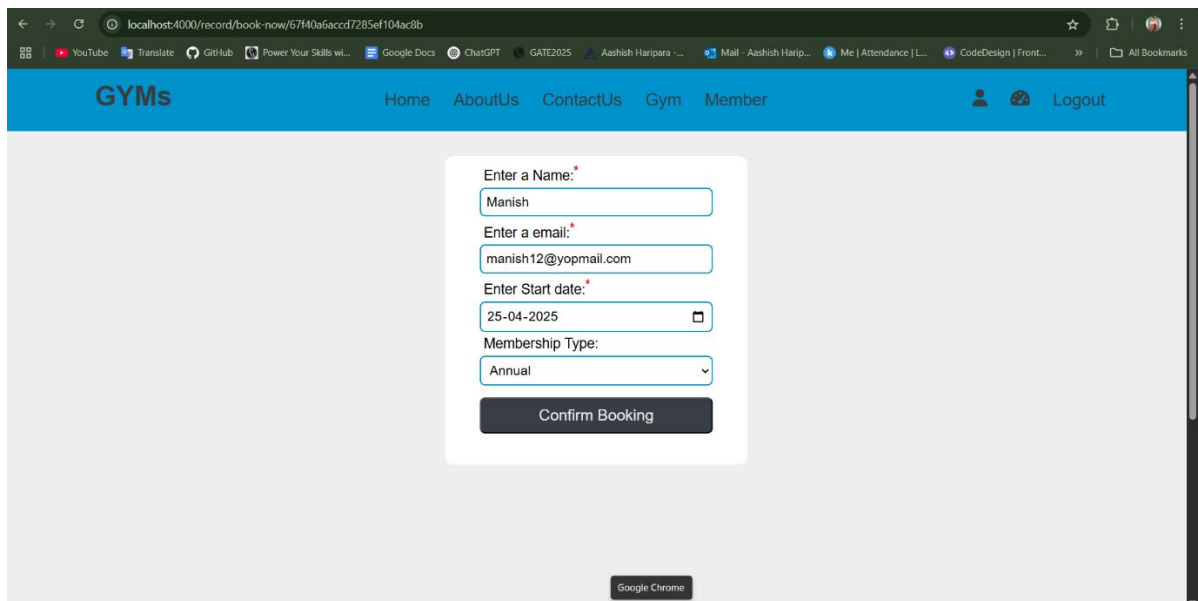


Fig. 9.4.1 Member Gym Booking Page

CHAPTER: 10

ADVANTAGES

10.1 Advantages

10.2 Limitations

10.1 Advantages:

- **Efficiency:** Enhances productivity by efficiently matching users with gyms based on location, facilities, and user preferences.
- **Time Management:** Helps users manage their workout schedules effectively by providing clear information on gym availability, classes, and peak times.
- **Resource Optimization:** Matches gym seekers with available gym slots, optimizing gym space and reducing idle times.
- **Prioritization:** Enables users to prioritize gym features like equipment availability, trainer expertise, or specific classes they are interested in.
- **Coordination:** Facilitates coordination between gym-goers and gyms, ensuring smooth booking processes and reducing conflicts.
- **Real-time Updates:** Provides real-time updates on gym availability, class schedules, and special offers, helping users make timely decisions.
- **Data-driven Insights:** Generates insights on gym popularity, user preferences, and utilization rates, aiding gyms in optimizing their services and offerings.
- **Reduced Errors:** Minimizes errors in gym selection and scheduling, enhancing user satisfaction and gym operational efficiency.
- **Adaptability:** Offers flexibility to adjust to user preferences, gym availability, and changing fitness trends, ensuring personalized recommendations.
- **Cost Savings:** Helps users find cost-effective gym options and enables gyms to optimize resource allocation, reducing operational costs.

10.2 Limitations:

Dependency on Data Accuracy: Relies heavily on accurate gym data regarding facilities, class schedules, and location details for effective recommendations.

- **Complexity:** Implementing and managing a sophisticated recommendation system may require training for users and gym administrators, potentially leading to adoption challenges.
- **Integration Challenges:** Integrating with existing gym management systems and databases can be complex, requiring compatibility checks and additional resources.
- **Scalability Issues:** As the user base grows or more gyms join the platform, scalability becomes crucial to handle increased data and user interactions effectively.
- **Security Concerns:** Handling sensitive user information like fitness preferences and

CHAPTER: 11

CONCLUSION AND FUTURE ENHANCEMENT

11.1 Conclusion

11.2 Future Enhancement

11.1 Conclusion

In conclusion, a **Gym Finder & Recommendation System** offers many advantages, with better efficiency, better time management, optimized resource utilization, improved coordination, and valuable functionalities for both users and gym owners. However, challenges such as data accuracy, system complexity, integration issues, and security concerns must be considered.

Despite these challenges, the benefits of a implemented Gym Finder & Recommendation System some the limitations, particularly in today's fitness-conscious and technology-driven world. A well-structured **Gym Finder & Recommendation System** can revolutionize how people discover and access fitness facilities, making gym selection easier, more personalized, and efficient.

11.2 Future Enhancement

Improves efficiency and user experience; future enhancements can focus on incorporating emerging technologies and refining existing functionalities. Some key areas for improvement include:

- **AI & Machine Learning Integration:** AI-driven recommendations can provide personalized gym suggestions based on user preferences, location, and fitness goals. Machine learning can analyse user behaviour to improve recommendations over time.
- **Advanced Analytics:** Enhanced data analysis can offer insights into gym popularity, peak hours, and user satisfaction, allowing gym owners to optimize services and users to take informed decisions.
- **Predictive Analytics for User Trends:** AI-based predictions can help gyms anticipate demand, adjust schedules, and offer dynamic pricing or discounts based on usage trends.
- **Mobile Accessibility:** A more robust mobile experience with features like real-time gym occupancy, instant booking, and virtual gym tours can improve user

engagement.

By continually improving and expanding the system, a **Gym Finder & Recommendation System** can provide even greater value to users and gym owners, helping people achieve their fitness goals efficiently while optimizing gym operations.

BIBLIOGRAPHY

a. Course Outcome

b. Books

c. Web Reference

a. Course Outcome

Sr. No.	CO statement	Marks % weightage	Chapter No.
CO-1	Undertake problem identification, formulation and solution	20%	CHAPTER 1 INTRODUCTION
CO-2	Design engineering solutions to complex problems utilising a systematic approach and team work	30%	CHAPTER 2 SYSTEM REQUIREMENTS CHAPTER 3 DAILY TASK
CO-3	Communicate with engineers and the community at large in written and oral forms	20%	CHAPTER 6 SYSTEM DESIGN CHAPTER 10 ADVANTAGES CHAPTER 11 CONCLUSION AND FUTURE ENHANCEMENT
CO-4	Demonstrate the knowledge and understanding of engineering and management principle and apply it to assigned project	30%	CHAPTER 4 FRONT END OF SYSTEM CHAPTER 5 BACK END OF SYSTEM CHAPTER 6 SYSTEM DESIGN CHAPTER 7 DATA DICTIONARY CHAPTER 8 TESTING

b. Books:

- **Beginning Node.js**
 - Author: Basarat Ali Syad

c. Web Reference:

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